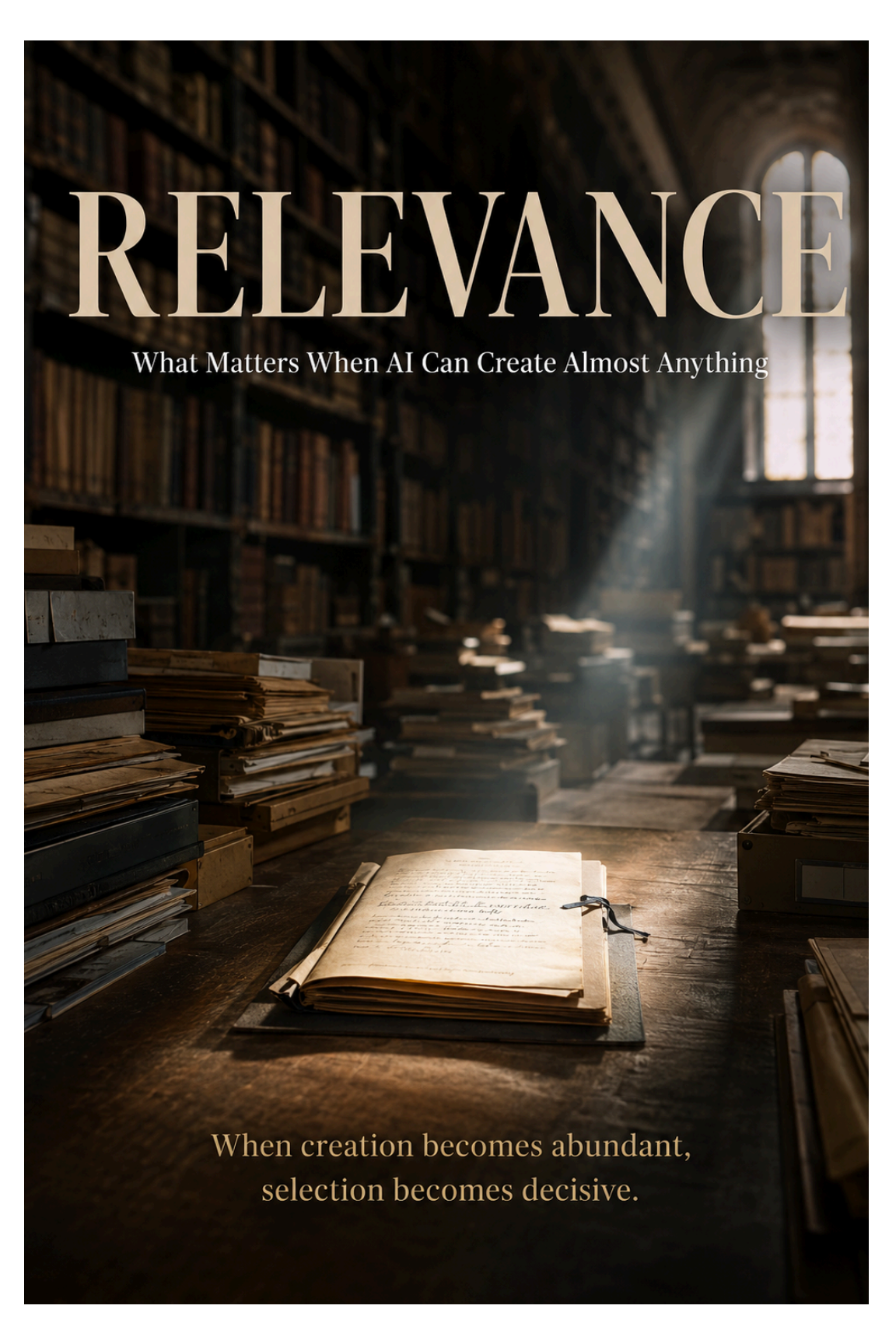




RELEVANCE

What Matters When AI Can Create Almost Anything

When creation becomes abundant,
selection becomes decisive.

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboaie. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

Outfinity Venture Studio: www.outfinity.ch

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1. When Creation Stops Being Scarce

A century ago, a person with an idea faced the problem of making it real. Writing a book required months or years. Testing a scientific hypothesis required scarce instruments and labour. Producing a film consumed capital before an audience could see the first finished version. Even generating a serious design alternative cost time. Those constraints were often unfair, but they gave intellectual production a built-in throttle. The world did not need a formal theory of what to inspect first because production itself limited how many candidates arrived.

Generative AI weakens that throttle. The change is not that machines suddenly create without cost or that human judgment becomes obsolete. The change is comparative: plausible candidates can now be produced much faster than they can be verified, understood, tested, funded, remembered, or culturally absorbed. The scarce object moves from candidate generation toward disciplined selection. This book begins from that movement and asks a deceptively simple question: what does it mean for something to matter when there can be almost infinitely many things to choose from?

The answer cannot be a universal score. Relevance is not a substance contained inside a paper, project, novel, film, design, or image. It is a relationship between an artifact and a purpose in a context at a time. The first chapter builds this grammar carefully because every later chapter depends on it.

1.1 The Reversal of Scarcity

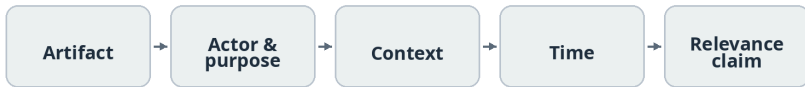
Imagine a research lab that asks an advanced model for promising directions in trustworthy artificial intelligence. By breakfast it has two hundred hypotheses, twenty experimental plans, a draft survey, candidate datasets, code for three baselines, and six possible grant concepts. Nothing in this scene is science fiction in the strong sense; it is simply the extrapolation of capabilities that already exist in partial form. The surprise is that abundance does not remove difficulty. It relocates it. Someone still has to decide which hypotheses deserve experiments, which evidence is strong enough to change direction, which project concept is more than fluent grant language, and which apparent novelty is merely a recombination of familiar terms. The scarce resource is no longer the ability to produce a candidate. It is disciplined attention applied to a field of candidates.

The same reversal appears outside science. A publisher facing ten manuscripts could afford to read them closely; a platform facing ten million generated books cannot. A design studio once paid heavily for each concept render; a generative system can now create thousands before lunch. A film producer historically performed selection before committing cameras, sets, actors, and distribution budgets; synthetic media can move a growing part of that cost after selection. In each case the old production bottleneck served as an accidental filter. It was unfair, wasteful, and exclusionary in many ways, but it also limited volume. When that filter disappears, societies need intentional selection mechanisms or they drown in plausible artifacts.

The economic intuition behind this reversal is simple but easy to underestimate. When the marginal cost of producing one more plausible draft falls toward zero, quantity stops being evidence of effort or value. A laboratory can generate more hypotheses than it can experimentally test; a studio can generate more visual concepts than it can responsibly compare; a company can generate more strategic plans than management can understand. The bottleneck becomes the cost of falsification, comparison, interpretation, and

commitment. AI therefore increases the value of good evaluative institutions even when it improves generation itself.

Figure 1 deliberately uses a linear representation even though the real relationship is richer. Its purpose is to force the evaluator to name the variables that ordinary language often hides: the artifact, the actor and purpose, the surrounding context, and the time horizon.



The claim changes when the relation changes.

Figure 1. Relevance is a relationship rather than an intrinsic property.

The last box is a claim, not a discovered substance. That wording matters because claims can be supported, challenged, scoped, and revised.

When creation becomes cheap, the difficult problem moves to selection.

Before generative artificial intelligence, most intellectual culture treated production as the scarce part. You had to find a researcher capable of formulating a hypothesis, a writer capable of producing a book, a designer who could imagine a solution, or a team able to transform an idea into a fundable project. Evaluation came afterwards. It mattered, but it was secondary to the scarcity of creation. That order is beginning to reverse. If a machine can generate thousands of hypotheses, sketches, scenarios, or project concepts in a day, the economic and epistemic question is no longer only "can we create?" It becomes "what deserves our attention?" Relevance becomes infrastructure.

The word sounds ordinary because we use it as if it described a property. We say that a paper is relevant, a project is relevant, or a film is relevant to its time. Yet relevance is not like mass or length. A result can be relevant to a clinician and irrelevant to a physicist; relevant to a decision in 2026 and decisive only in 2040; relevant to a Horizon Europe call and a poor fit for the ERC; relevant to a subculture and invisible to the mass market. Information science has emphasized the relational and contextual character of relevance for decades. Tefko Saracevic showed that the same term spans algorithmic, cognitive, situational, and motivational manifestations. In research impact assessment, Dobrow and colleagues arrive at the same conclusion from another direction: relevance cannot be assessed without knowing for whom, for what purpose, and at what time the assessment is being made.

This distinction also clarifies why ordinary information retrieval is not enough. Retrieval asks which items match a query or a user profile. Relevance in the broader sense asks whether an item matters for a decision, a community, a problem, a cultural conversation, or a future trajectory. A paper can be semantically close to a query and still be methodologically useless. A project can repeat every keyword of a European call while contributing little to the underlying need. A novel can match a reader's declared preferences and still be forgettable. Matching is one ingredient in relevance; it is not the whole meal.

Once relevance becomes infrastructure, errors become asymmetric. A false positive wastes attention or money on something that does not matter. A false negative can be worse: it can eliminate a scientific direction, artistic voice, or technical architecture before it has a chance to reveal its value. The cost depends on the domain and the time horizon. A recommendation engine may tolerate occasional weak suggestions. A frontier research fund should be far more concerned about systematic rejection of unfamiliar ideas. This is the first reason the book resists universal scores: the same numerical mistake can have radically different consequences in different institutions.

The practical question is therefore not how to automate taste. It is how to make selection more explicit, testable, plural, and corrigible than the informal filters it replaces. That requires a language for describing what exactly is being judged. The next section develops that language by separating relevance from the concepts that are most often confused with it.

1.2 A Relationship, Not a Score

A useful way to see the relational nature of relevance is to follow one artifact through several contexts. Consider a new method for detecting a rare failure mode in medical AI. For a clinical safety team deploying such systems, the method may be immediately relevant even if the paper describing it is modestly cited. For a theoretical machine-learning group it may be peripheral. For a regulator it may become highly relevant only after a public incident exposes the failure mode. For an investor it may matter only if regulation creates demand. The artifact did not acquire a new molecular property each time. What changed was the decision situation around it.

This is why the phrases 'high quality' and 'high relevance' should not be interchangeable. Quality describes properties internal to execution: rigor, coherence, robustness, craftsmanship, or technical competence. Novelty describes distance from prior work. Utility describes contribution to a goal. Impact describes effects that actually occur. Popularity describes attention or aggregate preference. Fit describes compliance with a local brief. Relevance is the argued connection among some of these properties and a particular need, actor, context, and time. It can use them as evidence without collapsing into any one of them.

A practical test for conceptual confusion is to ask whether the judgment could change while the artifact stays fixed. If a paper becomes urgent after a new epidemic, if a design becomes unusable after a regulation changes, or if an old film becomes newly central after a social event, then the property being discussed cannot be only intrinsic quality. Relevance carries a reference class and a time index

even when ordinary language leaves them implicit. Good evaluation makes those hidden variables explicit.

This book starts from a simple proposition: relevance is an argued relationship between an artifact and an evaluation situation. The artifact may be a project proposal, a paper, a dataset, an algorithm, a book, a film, or a design object. The situation includes an actor or community, a purpose, an institutional context, and a time horizon. Remove any of these elements and the score becomes easier to calculate and harder to defend.

The distinctions in the table are not semantic housekeeping. They prevent an evaluator from treating one convenient proxy as if it answered every question. The first column names a concept that is often collapsed into relevance; the second explains what it can legitimately contribute.

Dimension	Interpretation
Quality	How well the artifact is constructed or executed. Quality can support relevance, but it does not by itself say for whom or for what the artifact matters.
Novelty	Distance from what existed before. Novelty without value can be noise; value without novelty can still be important consolidation.
Utility	The capacity to help achieve a goal. It is central in design; in art it can be too narrow if interpreted instrumentally.
Popularity	Attention, consumption, or aggregate preference. It is a social signal, not a definition of relevance.
Impact	Observable effects produced after use. Impact may be delayed, distributed, or contingent, so it should not be treated as a simple synonym for relevance.
Fit	Compatibility with a local brief, call, or need. Strong fit can coexist with low novelty or low long-term value.

Table 1. Distinguishing relevance from neighboring concepts.

The practical discipline is to keep these dimensions separate until the decision policy explicitly combines them. A ranking can then be

debated in terms of weights and trade-offs rather than hidden assumptions.

A decisive distinction follows. Relevance is not identical to truth, quality, novelty, utility, popularity, or impact. A result can be true and irrelevant to the present problem. An idea can be new and useless. A work of art can be unpopular when it appears and central to a later generation. A project can fit a call perfectly and still be scientifically banal. A paper can accumulate citations because it is controversial rather than correct. When we compress all these dimensions into a single number, we do not merely simplify the world; we change the question.

The best initial representation is therefore a relevance dossier rather than a relevance score. The dossier states the referent of the evaluation, the available evidence, the alternatives considered, the uncertainty, and the conditions under which the verdict would change. A score may be derived for a local decision, but the structure that produced it should remain visible. This idea will return in every domain of the book.

The distinction matters because institutions create incentives around whatever they measure. If relevance becomes citation count, researchers learn to optimize citation-producing behavior. If relevance becomes call-language similarity, grant writers learn to mirror policy vocabulary. If artistic relevance becomes engagement, creators learn to maximize immediate emotional response. None of these behaviors is irrational; the system asked for them. The mistake lies in believing that the proxy remained a faithful description of value after it became a target. Goodhart's law is especially powerful in the AI era because generators can search proxy space much faster than humans ever could.

A scalar relevance score is therefore best treated as a local compression performed at the end of an evaluation, not as the evaluation itself. Before compression, the system should retain the target of judgment, the evidence, the alternatives, the uncertainty, and the perspectives that disagree. A project might score highly for societal need but only moderately for intervention plausibility. A

book might be stylistically competent but culturally derivative. A design might solve the brief beautifully while failing accessibility constraints. Keeping these dimensions separate makes disagreement intelligible and revision possible.

This leads to the core object proposed throughout the book: a relevance dossier. A dossier does not claim that numbers are useless. It says that numbers need provenance. If a score is later required for ranking, the dossier should make it possible to reconstruct how the score was produced, which assumptions mattered, which evidence was weak, and what new information would cause the verdict to change. Relevance, in this sense, is not a mystical property. It is a disciplined and auditable relationship.

1.3 A Vocabulary for What Matters

The transition from research to literature, design, film, software, policy, and visual art becomes possible once we identify a common grammar. Every relevance judgment contains an artifact, a reference space, an actor or public, a purpose or mode of engagement, a context, and a time horizon. In science, the reference space may be the state of the art and the actor may be a research community. In design, the reference space includes existing products and constraints, while the actor is a user or organization. In literature, the reference space is partly a tradition of texts and partly a cultural conversation. The grammar is stable even when the criteria change.

Within that grammar, at least five families of relevance recur. Epistemic relevance concerns what an artifact changes about knowledge: a theorem, dataset, experiment, explanation, or conceptual distinction. Instrumental relevance concerns what it helps someone accomplish. Experiential relevance concerns what happens in the encounter between a person and the artifact: understanding, emotion, surprise, identification, or reflection. Cultural relevance concerns how artifacts alter shared language, memory, norms, representation, or genre. Systemic relevance concerns the effect of selecting the artifact on a wider portfolio or ecosystem.

So far, research has been treated as a special domain. The next step is more radical: many of its evaluative concepts can be generalized to almost any intellectual artifact produced by AI. Not because a poem is a paper or a film is a grant proposal, but because all of them enter an ecology of selection in which something worth continuing must be distinguished from a large space of possibilities.

The common structure is surprisingly stable. There is an artifact. There is a reference space: the corpus of previous work, user needs, an artistic tradition, a design brief, a technical problem, a cultural conversation. There are evaluators or beneficiaries with preferences and values. There are consequences and a time horizon. What changes across domains is the weight of these components and the legitimacy of particular criteria.

In design, utility and satisfaction of constraints carry substantial weight. In research, epistemic credibility is non-negotiable. In literature, factual truth may be secondary to coherence, semantic density, and the power to reconfigure a reader's perspective. In visual art, a work can be relevant precisely because it does not optimize immediate preference. In film, commercial success, critical judgment, and cultural impact can diverge dramatically.

This variation is why a universal "AI quality metric" is a category mistake. We can have a common architecture, but not a common scoring function. The architecture should ask what kind of relevance is being evaluated, who is entitled to judge it, what evidence is admissible, and which losses we accept if we are wrong.

The same object can occupy very different positions along these dimensions. A technical standard may be culturally invisible but systemically crucial. A novel may be instrumentally useless in a narrow sense and yet culturally transformative. A teaching resource may contain no novel science but be highly relevant because it changes understanding for a specific learner population. A software package may have modest novelty yet enormous systemic relevance because thousands of other tools depend on it.

This multidimensional view also prevents a subtle category error in AI evaluation. Because the same multimodal model can generate

code, images, essays, and project proposals, it is tempting to build one generic 'quality judge' for all of them. Uniform generation technology does not imply uniform value. Correctness is a threshold in code but not in fiction. User satisfaction is central in product design but cannot define frontier science. Popularity is economically important in film distribution but can be orthogonal to cultural significance.

What can be shared across domains is therefore not the scoring function but the evaluation architecture. The system can always ask who the evaluation is for, what kind of relevance is legitimate, which constraints are non-negotiable, what evidence is available, which uncertainties remain, and what alternatives are being compared. This common structure allows transfer of methods without erasing the ontology of each domain.

The result of this chapter is deliberately modest: relevance is a disciplined claim, not an intrinsic score. It becomes stronger when the evaluator names the purpose, context, time horizon, alternatives, evidence, and uncertainty. That grammar is sufficient to expose many category errors before any sophisticated model is used.

Science is the next step because it gives us the largest existing body of evidence about what happens when institutions try to operationalize such claims at scale.

2. Research: The First Laboratory of Relevance

Research is the best place to study relevance because science has spent decades building institutions that try to distinguish what is rigorous, new, useful, influential, and worth funding. Those institutions are imperfect, but their failures are visible and extensively studied. Citations, peer review, impact case studies, grant panels, bibliometric indicators, societal-impact frameworks, and responsible-assessment movements form a large natural experiment in how societies try to decide what knowledge matters.

The chapter follows a result from production to consequence. A paper, dataset, method, or piece of software first exists as an output. It may then be noticed, reused, criticized, adopted, translated into practice, or ignored. Some of those pathways lead to scientific or societal effects; many do not. The distinction sounds obvious, yet evaluation systems repeatedly collapse output, attention, quality, and impact into one another because the early signals are easier to count.

Science therefore contributes two lessons to the wider problem of AI-era relevance. First, useful evaluation needs multiple dimensions and explicit time horizons. Second, every metric becomes dangerous when institutions forget what it was originally capable of measuring.

2.1 From Outputs to Effects

Science provides the richest laboratory for studying relevance because it has spent more than a century building institutions that must choose under scarcity. Journals choose papers, funders choose proposals, universities choose people, governments choose programmes, and societies later ask whether any of those decisions produced knowledge or benefit. The resulting machinery is imperfect, but it has generated a vocabulary that other domains now need: peer review, bibliometrics, impact pathways, contribution, productive interaction, qualitative case studies, field normalization, and responsible metrics.

The earliest quantitative temptation was understandable. Citations are countable, visible, and connected to scientific use. A paper cited by many subsequent papers has clearly entered the scientific conversation. Yet the meaning of a citation is ambiguous. Researchers cite work they build on, criticize, review, ritualistically acknowledge, or use as a standard method. Citation cultures differ sharply across fields and over time. A slowly developing mathematical result and a fast-moving biomedical method do not inhabit the same temporal ecology. This is why modern bibliometrics normalizes by field and publication age and why even normalized citation impact is best interpreted as one signal of scientific influence rather than a general measure of relevance.

The European Commission's Key Impact Pathways are one institutional example of this shift. They attempt to follow scientific, societal, and economic/technological effects over short, medium, and longer horizons rather than treating publication output as the final state of a project. The framework is imperfect, but conceptually important: it treats impact as a temporal chain with different indicators at different stages. That is closer to causal reasoning than to scoreboard thinking.

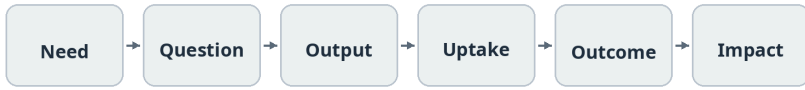
Research is the most useful starting point for a general theory of relevance, not because science has solved the problem, but because it has been forced to turn an intuition into an institutional process. Funding agencies, universities, and public programmes must decide

which projects deserve resources and which results demonstrate value. Under that administrative pressure, bibliometrics, impact studies, rubrics, peer review, theory of change, productive interactions, and movements for responsible research assessment emerged.

A foundational contribution is the salience-credibility-legitimacy model developed by Cash and colleagues. Knowledge that is useful for decision-making must be salient, meaning that it responds to a real need; credible, meaning that it has an adequate epistemic basis; and legitimate, meaning that it is produced and evaluated through a process that stakeholders can regard as fair. Relevance appears here under the name salience, but the framework is stronger than a synonym: it shows that a relevant but non-credible result can be dangerous, while a credible but non-salient result can remain without effect.

Research Quality Plus, developed by IDRC, operationalizes a related idea by assessing rigor, legitimacy, importance, and positioning for use. SIAMPI shifts attention from final impact, which may appear years later, toward productive interactions between researchers and social actors. ASIRPA reconstructs impact pathways through case studies. The United Kingdom's REF uses thousands of impact case studies to assess extra-academic effects without pretending that a single metric can capture their meaning.

The diagram separates production from uptake and effects. Many assessment mistakes happen because indicators from the left side of the pathway are interpreted as if they already demonstrated the right side.



Output is evidence of work; impact is evidence of effects.

Figure 2. A research output is only one stage in a longer pathway.

The further the evaluator moves toward outcomes and impact, the more causal attribution becomes difficult. That uncertainty should be represented rather than hidden behind a single cumulative score.

As research policy began to demand evidence of societal value, evaluation moved beyond publications. The productive-interactions approach associated with SIAMPI asks whether researchers and social actors engage in exchanges that make later uptake possible. Theory-of-change approaches ask evaluators to reconstruct a causal sequence from activities to outputs, outcomes, and broader effects. ASIRPA uses case-based impact pathways to understand mechanisms and contribution. The United Kingdom's Research Excellence Framework institutionalized impact case studies at enormous scale. These approaches are expensive compared with counting papers, but their cost buys something important: context.

Research Quality Plus, developed in the international-development context, also matters because it refuses to pretend that rigorous work automatically becomes useful. It distinguishes dimensions such as rigor, legitimacy, importance, and positioning for use. This is close to the salience-credibility-legitimacy framework developed by Cash and colleagues for knowledge systems. A piece of knowledge can answer the right question yet be untrustworthy; it can be methodologically strong yet irrelevant to the actors who must act; it can be both credible and salient yet produced through a process stakeholders see as illegitimate. Relevance is therefore entangled with, but not reducible to, epistemic and institutional quality.

The historical movement from metrics to pathways contains the first major lesson for AI-generated artifacts. When the connection between an artifact and its effects is indirect, evaluation must model the intermediaries. A paper does not jump directly into social impact. A design does not jump directly into adoption. A film does not jump directly into cultural memory. Between production and consequence lie institutions, users, complementary technologies, distribution channels, norms, incentives, and chance. Any general science of relevance needs to represent those bridges rather than merely count endpoints.

Once a project ends, evaluators finally have artifacts rather than promises. That seems simpler, but it introduces a new problem: research outputs are heterogeneous. A paper, dataset, piece of software, patent, standard, benchmark, trained model, clinical protocol, negative result, and policy guideline leave different traces in the world. Treating all of them as if citations were the universal currency of relevance systematically undervalues infrastructure and practice-oriented work.

Publications are the easiest case because mature bibliometric systems already exist. Field- and time-normalized citations can show whether work has been used within the scientific literature. Citation context can go further by distinguishing background mention from methodological dependence or critique. Yet even an ideal citation system measures scientific uptake, not necessarily truth or societal value. A highly cited method may later be discovered to be flawed. A foundational result may be used implicitly without being cited. A paper addressing a narrow but crucial safety problem may be extremely relevant to a small community while remaining globally obscure.

Once research has produced results, assessment seems easier. We now have publications, datasets, software, prototypes, patents, standards, guidelines, and perhaps users. In reality, the problem has merely changed form. Ex post relevance is richer in evidence, but also more vulnerable to confusion between visibility and value.

Citations are useful for seeing whether a result has entered the metabolism of science. Citations normalized by field and time are more informative than raw counts. Yet citation can signal positive use, criticism, a standard method, controversy, or simply a position within a citation-dense field. Altmetrics add Mendeley, news, social media, and other traces, but the literature repeatedly shows that these sources measure different behaviors. A tweet and a mention in a guideline are not two units of the same substance.

For software and data, relevance can often be observed more directly through reuse, dependencies, integrations, recurring downloads, external contributions, or use in pipelines. For patents and technologies, licensing, standards, integration into products, and subsequent investment may provide signals. For policy, citations in official documents and interviews with decision-makers can reconstruct influence. No domain, however, has a universal proxy.

Sleeping beauties are the temporal warning. Some results have little impact for years and become central only when new technologies or questions appear. A system that judges too early optimizes for proximal results. A system that judges too late cannot guide current allocation. The answer is not a single time window but evolving relevance profiles.

Software and datasets require different evidence. Relevance may appear through package dependencies, active installations, reuse in published studies, integration into production systems, forks, issue activity, benchmark adoption, or the cost that others would incur if the resource disappeared. A small open-source library deep in a dependency chain can be more infrastructurally relevant than a popular application. For datasets, reuse across independent studies and tasks may matter more than download counts. For benchmarks, relevance can become paradoxical: success may eventually require replacing the benchmark because the field has overfit to it.

Patents and standards create yet another topology of impact. A patent can be cited by later patents without becoming a product. A technical contribution can be incorporated into a standard and then disappear from ordinary academic visibility precisely because it has

become infrastructure. Policy guidelines may matter through adoption by agencies rather than citations. Negative results can prevent costly dead ends without generating glamorous output. In each case the evaluator needs a domain-specific theory of use.

This heterogeneity is a strong argument for relevance dossiers. The dossier begins by identifying what kind of artifact is being assessed and what evidence would count as meaningful uptake for that artifact. It can then combine quantitative traces with qualitative reconstruction. The objective is not to give every output its own incomparable narrative, but to avoid using evidence simply because it is available rather than because it measures the construct of interest.

2.2 Responsible Assessment and the Metric Trap

The reform of research assessment over the last decade is, in part, a reaction to what happens when convenient indicators harden into institutional truth. DORA argues against using journal-level metrics such as the Journal Impact Factor as substitutes for the quality of individual contributions. The Leiden Manifesto insists that quantitative evaluation should support qualitative expert assessment and be sensitive to field differences, local missions, and data quality. CoARA extends this agenda across Europe by making qualitative judgment central while permitting responsible quantitative support. DORA's 2024 guidance distills the same idea into five requirements for indicators: they should be clear, transparent, specific, contextual, and fair.

The important point is not that expert judgment is pure while metrics are corrupt. Human review has its own biases, status effects, fatigue, disciplinary conservatism, and inconsistency. Rather, different forms of evidence fail differently. A citation metric can be audited mathematically but may measure the wrong construct. A panel can interpret context but may reproduce social bias. A case study can reveal causal mechanisms but is expensive and difficult to compare. A good assessment system should therefore combine

heterogeneous signals in ways that preserve their limitations instead of averaging them into apparent certainty.

The reform movement has also become institutional rather than rhetorical. DORA's 2024 guidance on quantitative indicators emphasizes clarity, transparency, specificity, context, and fairness; CoARA's agreement places qualitative judgment at the center of research assessment while allowing responsible quantitative support. The point is not anti-metric romanticism. It is recognition that metrics are measurements with validity domains, not portable currencies of merit.

These frameworks share one lesson: impact is a process, not an instantaneous property of a publication. Between a result and a change lie users, institutions, policies, complementary technologies, barriers, alternatives, and chance. For this reason, the idea of pure attribution - project X caused exactly Y percent of the impact - is often fictitious. What can be reconstructed more defensibly is contribution: the project was necessary, facilitating, accelerating, or one among several sufficient causes in a mechanism.

Research evaluation has accumulated a toolbox rather than a single instrument. Each method illuminates a different stage of the pathway from work to effect, and each has characteristic blind spots.

Method family	What it can realistically tell us
Bibliometrics	Useful for scientific influence and diffusion through the literature; weak as a direct measure of societal value.
Altmetrics	Fast signals of attention and circulation; heterogeneous constructs that are easy to confuse with impact.
Productive interactions	Observe interactions that make uptake possible before final impact becomes visible.
Case studies	Reconstruct context, mechanisms, and contribution; provide depth at high cost and with lower mechanical comparability.

Method family	What it can realistically tell us
Theory of Change	Makes the need–output–outcome–impact chain and its assumptions explicit; can become a formal exercise if it is not tested.
Responsible metrics	Uses indicators in support of judgment, with normalization, transparency, and attention to incentive effects.

Table 2. What common research-assessment methods can and cannot tell us.

The table should therefore be read as a map of evidential roles, not as a menu from which the evaluator chooses one winner. Mature assessment triangulates across methods when the decision stakes justify the cost.

Bibliometrics remains valuable, but in a more modest role than the mythology of indicators suggests. Citations normalized by field and time can show influence within the scientific system; altmetrics can show attention or circulation; policy-document citations can indicate a bridge toward decision-making. None of them, by itself, is relevance. DORA, the Leiden Manifesto, and CoARA converge precisely on this point: assessment should rely primarily on contextual judgment, responsibly supported by indicators rather than replaced by them.

This becomes even more important in the AI era. If a system can optimize text against visible metrics, every simple indicator becomes a target. Goodhart's law is not a design accident; it is a property of evaluation when incentives are strong. Relevance systems should therefore expose their mechanisms, compare heterogeneous evidence, and preserve legitimate disagreement.

AI intensifies the problem because it makes the evaluation target adaptive. Once applicants know that a machine checks semantic alignment with call text, they can generate near-perfect alignment language. Once creators know that an aesthetic model predicts audience preference, they can optimize images against that model. Once researchers know that an automated novelty score rewards semantic distance, they can manufacture distance without

substantive contribution. The evaluator becomes part of the environment. Its criteria alter the population it later measures.

This performative loop suggests that evaluation systems should themselves be evaluated experimentally. We should ask whether a metric remains informative after people or generator agents optimize against it, whether different groups can game it equally, whether it narrows the range of accepted work, and whether it suppresses unusual but later valuable artifacts. These are not secondary governance questions. They are questions about construct validity under strategic pressure.

The most mature research-assessment movements therefore point toward a principle that will recur in every later chapter: use metrics as sensors, not sovereigns. A sensor can contribute evidence about one dimension. It does not decide what the institution ought to value. The decision layer must remain explicit about purpose, trade-offs, uncertainty, and acceptable error. That is exactly the architecture required when AI moves from producing intellectual artifacts to evaluating them.

2.3 Time, Contribution, and Delayed Relevance

Time changes relevance in ways that static evaluation rarely captures. The literature on 'sleeping beauties' describes work that receives little attention for years and then becomes influential when another discovery, technology, or problem makes it newly usable. The phenomenon matters beyond bibliometrics. A software architecture may appear unnecessary until systems scale. A dataset may become valuable when a new model class can exploit it. A speculative artistic work may acquire cultural force after events make its language newly intelligible.

Delayed relevance exposes the weakness of retrospective storytelling. Once a result becomes important, institutions are tempted to reconstruct a smooth path in which the original project obviously led to the later impact. Real causality is usually messier. Outcomes emerge from networks of researchers, funders, complementary inventions, adopters, regulation, infrastructure, and

accidents. Impact assessment is therefore more defensible when it speaks of contribution rather than claiming precise attribution that the evidence cannot support.

Another problem is the level of aggregation. The relevance of a publication is not the same as the relevance of a project, and the relevance of a project is not the same as the relevance of a portfolio. A good programme may fund individually risky projects with uncertain results precisely to create option value at the portfolio level. When every element is evaluated with the same rubric, the diversity the system needs can disappear.

This lesson will become crucial for AI. If generation is cheap, we do not want an evaluator that always selects the artifact with the highest average score. We want a system that knows when to preserve a promising anomaly, a minority direction, or an idea that is difficult to compare. Relevance is sometimes local and sometimes systemic.

Ex post relevance must also recognize negative results. A rigorous experiment that disproves an important hypothesis can be relevant even if it produces no product, patent, or adoption. A clean dataset showing that a research direction does not work can save others years of effort. Impact systems built exclusively around positive uptake tend to erase this value. In a more automated science, where millions of hypotheses may be tested, information about failure becomes infrastructure: relevance no longer means only "what worked," but also "which part of the search space can we now eliminate with justification."

A second difficult case is infrastructure. Software, tools, benchmarks, ontologies, and databases can have enormous relevance without looking spectacular in bibliometrics. They create options for future research. Dependency graphs, reuse, external contributions, pipeline integration, and avoided cost can be more informative than citations. This shift prepares the transition to AI artifacts: if generative systems increasingly produce reusable components, their value must also be measured by what they make possible for others, not only by direct consumption.

Contribution analysis asks what role the result played in a mechanism. Was it necessary? Did it accelerate adoption? Did it reduce uncertainty? Did it provide a missing component? Did it change the decision of a particular actor? These questions can be supported by process evidence such as version histories, standards documents, citations in policy drafts, interviews, procurement decisions, or dependency graphs. They are harder to reduce to one metric, but they are closer to the causal meaning of relevance.

Longitudinal evaluation also changes how we should judge the evaluator. If a funding panel labels a project highly relevant, that prediction should be stored and revisited. If an AI system predicts that a paper will matter to a field, the later trajectory should be used to calibrate it. If a cultural platform predicts that an unusual book has durable relevance, persistence over years matters more than first-week engagement. Evaluation becomes a forecast that can itself be scientifically tested.

The right conclusion is not that we must wait decades before making any decision. Decisions cannot wait. The conclusion is that relevance should be timestamped. An *ex ante* judgment, an immediate *ex post* judgment, and a ten-year retrospective answer different questions. A mature system preserves all three rather than allowing later success or failure to rewrite what was reasonably knowable at the time.

Research assessment shows both the necessity and danger of proxies. We need signals because effects often arrive late, yet early signals can become targets and distort behaviour. Responsible assessment is therefore not the rejection of measurement but the refusal to forget measurement error and causal distance.

The problem becomes harder when the work has not happened yet. Proposal review asks institutions to infer future relevance from arguments, teams, and plans, which is where the next chapter begins.

3. Judging Work Before It Exists

A completed research result can at least be inspected. A proposal is different: it asks evaluators to judge a future that exists only as argument, evidence, team capability, and plan. The applicant describes a problem, claims that something important remains unknown, proposes a mechanism for learning, and asks an institution to believe that the expected information or capability is worth the cost. Proposal review is therefore relevance under unusually high uncertainty.

This is why fluent grant writing can be misleading. A proposal can mention every fashionable objective and still be scientifically weak; it can be scientifically brilliant and still be irrelevant to a targeted programme; it can be feasible but strategically redundant; it can be risky in a way that should be rewarded in a frontier fund and rejected in a deployment programme. The evaluator is not asking one question but several questions whose weights depend on the institution.

The chapter treats peer review as a measurement system rather than a ritual of expert authority. That framing lets us ask how much signal it contains, where error enters, how identity and reviewer assignment matter, when randomization is more honest than false ranking precision, and what role AI can play without becoming a new opaque gatekeeper.

3.1 A Proposal Is a Hypothesis About the Future

A completed paper can at least be inspected. A proposal is different: its most important claims concern events that have not happened. The reviewer is asked to infer whether a method will work, whether a consortium can execute it, whether users will care, whether a technology will mature, and whether effects will propagate beyond the project. In other words, proposal review is a form of disciplined counterfactual reasoning. The object under evaluation is not only a document; it is a model of a possible future.

The strongest proposals make this model inspectable. They show why a problem exists, what the state of the art fails to do, why the proposed intervention could change that situation, which resources and dependencies are required, and what observable results would count as progress. Weak proposals often substitute importance for intervention logic. Climate change is important; that does not make every climate project relevant. AI safety is important; that does not establish that a particular benchmark, platform, or consortium is the best use of resources. The missing step is comparative reasoning about alternatives.

Evaluating a proposal is an epistemically strange act. The reviewer must judge a world that does not yet exist. The result, adoption, and impact cannot be observed; only arguments about them can be evaluated. Ex ante relevance is therefore not anticipated impact in a literal sense, but the quality of a hypothesis about the future.

A strong project connects a demonstrated problem to a knowledge gap, explains why existing alternatives are insufficient, proposes a mechanism through which activities will produce results, and identifies who might use those results. A weak project jumps directly from "the problem is important" to "the impact will be major." What is missing between those statements is exactly what an evaluator should be able to challenge: the causal chain.

Ex-ante evaluation is strongest when it distinguishes what can be judged from the proposal itself from what remains genuinely unknowable. The table summarizes dimensions for which reviewers can seek evidence without pretending that future outcomes have already occurred.

Dimension	What can be assessed ex ante
Problem importance	Is there a demonstrable need, contradiction, or limitation, and for whom does it matter?
Novelty	In what sense does the proposal exceed the state of the art, and what comparison makes the novelty claim testable?
Plausibility	Are the mechanism, methodology, and resources sufficient for the promised results?
Strategic fit	Does the project contribute to concrete programme objectives, or merely reuse their vocabulary?
Pathway	Is there a coherent path from results to users, outcomes, and broader effects?
Uncertainty	Which assumptions could fail, and how would the verdict change if they are false?

Table 3. Dimensions that can be assessed before a project exists.

The final funding decision may compress these dimensions, but the underlying record should preserve them. Otherwise the institution cannot learn whether it systematically overvalued feasibility, undervalued novelty, or confused applicant reputation with project credibility.

Grant peer review remains noisy. The literature finds limited reviewer agreement, effects of reviewer characteristics, reputational influences, and calibration problems across panels. A classic study of more than 130,000 NIH grants found that better peer-review scores were associated with better subsequent outcomes, showing that the process contains signal. The existence of signal does not remove the noise. Other work shows that changing reviewers can materially change funding probability, and that ranking may be more reliable than rating on ordinal scales.

Results published in 2026 make the problem even more interesting. An analysis of 37,073 ERC applications identified homophily: reviewers who share the applicant's country of work or citizenship tend to assign higher scores. Another study of how reviewers combine criteria shows that commensuration changes roughly 29 percent of final scores and that technical merit implicitly receives about twice the weight of societal relevance, even when the funder explicitly defines several dimensions. Criteria are therefore not merely read; they are reinterpreted by evaluators.

This is why a good *ex ante* assessment needs a counterfactual baseline. What happens if the proposal is not funded? Which competing approaches already exist? Which parts of the problem are likely to be solved anyway? Which outcomes depend uniquely on the proposed project, and which depend mainly on external developments? These questions protect evaluators from a common rhetorical move: describing a very important problem and then allowing some of that importance to spill automatically onto the proposed intervention.

Novelty adds another layer of uncertainty. The most original ideas are often the hardest to compare because their nearest precedents are weak or absent. In August 2026, Diogo Machado reported a small-sample study of 139 Austrian Science Fund proposals in which GPT-4o-based selection agreed with human funding decisions in roughly sixty percent of cases, while disagreement was systematically associated with human-rated novelty. More novel proposals were more likely to be funded by humans but not by the LLM. The sample is too small to justify a universal claim, but the result is a warning about a plausible failure mode: models trained on historical regularities may mistake familiarity for merit.

An AI assistant for proposal review should therefore be strongest where the problem is decomposition, traceability, and stress testing. It can identify unsupported claims, compare the proposed state of the art with literature, map objectives to methods and work packages, detect missing baselines, ask what would falsify a pathway, and generate adverse scenarios. Those tasks reduce cognitive load without pretending that the model possesses privileged access to the

future. The reviewer remains responsible for the value judgment, but the structure of the judgment becomes more inspectable.

3.2 Peer Review Is a Measurement System

Peer review is often discussed as if it were a ceremony of expert wisdom. It is more useful to treat it as a measurement system. Like any measurement system, it has signal, noise, calibration problems, and systematic bias. Reviewers differ in severity; panels differ in norms; disciplinary expertise shapes what looks plausible; reputations can leak into judgments; and small wording changes can affect interpretation. The right question is not whether peer review is 'objective' but whether its error distribution is acceptable for the decision being made.

Recent evidence from the European Research Council makes the point concrete. Fuchs-Schundeln and Holub analyzed 37,073 ERC applications from 2019 to 2023 and exploited the fact that the same proposal receives multiple reviews. Reviewers who shared an applicant's country of work or citizenship tended to award higher scores, even after accounting for proposal-specific quality and reviewer-specific leniency. The study does not imply that ERC evaluation is generally unreliable; it demonstrates that who evaluates can still matter in a process designed around scientific excellence.

This is where the temptation to use AI appears: if humans are inconsistent, ask an LLM for a score. It is precisely where caution is most necessary. In August 2026, a Scientometrics paper reported a signal of GenAI bias against scientific novelty in a small sample of proposals. The finding needs replication, but the direction is plausible: models trained on historical regularities may confuse stylistic plausibility and conceptual familiarity with merit. An evaluation system that penalizes ideas because they are hard to classify could automate conservatism at the very moment it claims to reduce it.

The language of measurement helps because it makes familiar concepts available: calibration, inter-rater reliability, measurement

error, construct validity, and sensitivity to the instrument. A panel can be highly reliable and still measure the wrong construct; it can also measure the right construct noisily. This distinction is crucial when evaluating reforms. Increasing agreement is not automatically progress if the new procedure makes everyone consistently conservative.

The conclusion is not to remove AI from assessment, but to change its role. AI is better suited to decomposition than sovereignty. It can extract claims, check whether claims have evidence, compare objectives with the call, identify dependencies, construct adverse scenarios, and show a reviewer where a score is sensitive to assumptions. In other words, it can turn an impression into a decision dossier.

There is another difficulty. The evaluator is not comparing the project with an empty future, but with a portfolio of alternatives. Saying that a problem is important does not prove that a particular intervention is the best use of resources. *Ex ante* relevance becomes stronger when the proposal contains a real comparator: what happens if the project is not funded, which technical or scientific alternatives exist, which part of the problem remains unsolved, and what options the result would open. In economics this resembles opportunity cost; in science, it is a discipline of the counterfactual. Without a comparator, the reviewer judges promises in a vacuum and becomes vulnerable to rhetoric.

This observation explains why ranking and rating are not equivalent. An absolute score of 4.5 out of 5 implies that the reviewer possesses a stable scale. In practice, many grants of similar quality are difficult to separate, and panels implicitly build their standards while comparing proposals. Studies comparing ranking with rating suggest that direct ordering can reduce some panel sensitivity. But ranking introduces its own problem: a good proposal can fall simply because it entered an exceptional cohort. A relevance system should therefore preserve both absolute judgments by dimension and pairwise or portfolio comparisons.

Where differences are smaller than evaluation noise, partial lotteries are not absurd. They are an institutional acknowledgment of epistemic limits. If ten projects clearly exceed quality thresholds but their ordering is not robust to reviewer changes, a controlled random selection may be more honest than fictitious numerical precision. For AI this matters: automation should not manufacture resolution that the evidence does not support. A mature evaluator must be able to say not only "A is better than B," but also "the available information does not justify ordering A above B."

Once peer review is viewed as measurement, several institutional practices become easier to understand. Calibration sessions are attempts to align scales. Multiple reviewers are repeated measurements. Panel discussion is an aggregation procedure. External referees add specialized instruments. Ranking and rating are different measurement designs: rating asks reviewers to locate proposals on an implicit scale, while ranking asks them to compare alternatives directly. In dense regions where many proposals are nearly indistinguishable, the choice of aggregation procedure can change outcomes more than the written criteria suggest.

Partial lotteries are controversial because they make randomness explicit, but that can be more honest than hiding randomness inside decimal scores. If a set of proposals clearly exceeds a quality threshold and their ordering is not stable under reviewer substitution, a lottery acknowledges an epistemic limit. The same principle should constrain AI systems. Automation should not manufacture precision merely because a model can emit a number with three decimals. A mature evaluator should be able to return an equivalence class, a confidence interval, or a statement that the evidence does not justify ordering A above B.

The final step is portfolio reasoning. Funders do not actually need the ten highest-scoring isolated proposals; they need a portfolio that advances a mission under uncertainty. Two projects may each be excellent yet redundant together. A slightly weaker project may add a missing method, population, region, or high-risk hypothesis. This shifts relevance from an object-level property to a system-level relation. It also creates one of the strongest future roles for AI:

comparing hundreds or thousands of proposals simultaneously to reveal overlap, complementarity, and risk concentration that no individual reviewer can easily see.

3.3 What Funders Are Really Deciding

A funding decision is often described as if it were a verdict on scientific quality. In practice it is a resource-allocation decision under uncertainty. The distinction matters. A review panel may believe that two proposals are both excellent and still need to choose between them because the budget cannot support both. A programme manager may prefer a proposal that covers an otherwise neglected capability even if another proposal has a slightly stronger conventional score. A public funder may need to consider whether a project advances a mission that private capital is unlikely to finance. Once scarcity enters the picture, evaluation is no longer only epistemic judgment; it is also portfolio construction.

Different funders make this visible in different ways. The U.S. National Science Foundation evaluates intellectual merit and broader impacts, while also allowing portfolio considerations such as transformative potential, capacity building, and programme objectives to inform final decisions. The National Institutes of Health reorganized most research-project peer review for applications due from January 2025 around three questions: how important the research is, how rigorous and feasible the approach is, and whether the expertise and resources are sufficient. The reform is revealing because it tries to reduce reputational bias by treating investigator and environment more as conditions of adequacy than as free-standing prestige scores. The object of judgment becomes the proposed work in context rather than the aura surrounding the applicant.

Some agencies go further and explicitly recognize that ranking precision collapses near the funding line. The Swiss National Science Foundation uses a Bayesian ranking procedure to account for differences in reviewer severity and uncertainty; when proposals at the cut-off are judged to be of equal scientific quality, a lottery may

be used. New Zealand's Health Research Council has long used random ordering among Explorer Grant applications that meet a threshold of transformative potential and viability. These mechanisms are not admissions of irrationality. They are acknowledgements that a decimal rank can imply a level of knowledge the evidence does not support.

The empirical literature gives reasons for this humility. Li and Agha's large study of NIH R01 awards found that better peer-review scores among funded grants predicted stronger subsequent publication, citation, high-impact publication, and patent outcomes. Peer review therefore contains real signal. But a signal is not omniscience. Other studies show low inter-reviewer agreement, sensitivity to panel composition, and difficulty separating close proposals. A system can be useful on average while still being unreliable at precisely the boundary where the decision is hardest.

The August 2026 ERC homophily study sharpens the point. Using more than 37,000 applications, Fuchs-Schundeln and Holub found that reviewers sharing an applicant's country of work or citizenship gave higher scores, with smaller same-gender effects. Their within-proposal design controls for proposal-specific quality and reviewer leniency, which makes the result more informative than a simple comparison of success rates. The implication is not that peer review is worthless; it is that reviewer assignment is part of the measurement system. Who reads a proposal can alter the observed signal.

Artificial intelligence does not automatically remove these problems. It changes their geometry. A model can be more consistent than an exhausted panel, but consistency around the wrong proxy is not fairness. In a 2026 perturbation study of EPSRC proposals, section-by-section LLM review detected some deliberately introduced weaknesses better than single-pass or multi-persona approaches, yet the models were much better at alignment and compliance failures than at holistic clarity. A separate 2026 analysis of historical Swedish grant data found only weak positive correlations between LLM scores and expert scores. These are early

results, but they already argue against treating a language model as an invisible replacement reviewer.

The strongest warning comes from Machado's August 2026 study of 139 Austrian Science Fund proposals. In that sample, a GPT-4o-based selection procedure agreed with human funding decisions in roughly three fifths of cases, but disagreement was structured rather than random. Proposals that human experts rated as more novel were disproportionately likely to be funded by humans and rejected by the LLM. The sample is small and the design cannot establish a universal mechanism, yet the failure mode is strategically serious: an evaluator trained on past language may systematically under-select ideas that depart furthest from the past.

This suggests a better division of labour. Machines can check internal consistency, trace claims to evidence, compare a proposal with literature, test whether milestones follow from objectives, detect budget or timeline anomalies, and expose where reviewers disagree. Humans can remain responsible for value-laden judgments, unusual novelty, programme intent, and decisions whose consequences cannot be reduced to observable historical data. The most promising system is therefore not human versus AI. It is an auditable interaction in which each side is used where its error profile is visible and contestable.

Once the decision is understood as allocation rather than pure ranking, one more concept becomes necessary: option value. A proposal can be weak on expected immediate return and still be worth funding because it opens a new search space, builds a capability, or creates information that changes future decisions. Frontier programmes often need a deliberate budget for such options. Otherwise, every rational local decision can add up to a globally conservative portfolio.

The central lesson is that a funder should be able to say not only why a proposal scored well, but what role it is meant to play in a portfolio, what uncertainty remains, and what kind of mistake the process is trying hardest to avoid. That is a richer idea of relevance than 'high score equals good project', and it prepares the ground for

the European programmes, where institutional purpose becomes even more explicit.

A further improvement is to state the loss function of the funding process in ordinary language. A mission-oriented programme may regard a false negative on a transformative idea as more damaging than a false positive on an ambitious experiment that later fails. A near-market instrument may rationally make the opposite trade-off because capital, deployment commitments, and time-to-market matter more. Once this is explicit, reviewer disagreement becomes easier to interpret: reviewers may not be disagreeing about facts at all; they may be assigning different costs to the same possible mistake.

This perspective also clarifies why ex-post success cannot be the sole validation target for grant review. Research portfolios contain genuine uncertainty, and a good decision can lead to a failed project just as a poor decision can occasionally lead to an accidental success. Validation should therefore examine calibration across classes of proposals, the diversity of funded bets, whether predicted risks actually materialized, and whether the process learned from misses. The relevant question is not whether evaluators predicted every future correctly, but whether their judgments were defensible given the evidence available at the time and improved when new evidence arrived.

Grant review is neither a magic filter nor random theatre. It contains real expert signal and real measurement error. The mature institutional response is to preserve the signal while making uncertainty, reviewer effects, portfolio logic, and the cost of false negatives more explicit.

Europe provides an especially useful next case because it makes several objective functions visible within one funding ecosystem.

4. Europe as a Relevance Machine

European research funding is an unusually rich case study because different instruments define relevance in visibly different ways while operating within the same broad policy ecosystem. Horizon Europe collaborative calls ask proposals to connect excellence with expected outcomes and impacts. The European Research Council protects frontier-research autonomy by treating excellence as the sole criterion. The European Innovation Council adds market creation, technological risk, investment logic, and need for Union support. None of these systems is simply 'more relevant' than the others; they solve different allocation problems.

This chapter reads European evaluation criteria not as grant-writing advice but as institutional philosophy. Each rubric reveals what the programme believes public money is for. The interesting question is not how to repeat the rubric's vocabulary. It is how to reconstruct the causal and strategic model behind it: which problem is being addressed, why European action is justified, what outcome is expected from the proposed work, and how the project changes a larger portfolio rather than merely satisfying a form.

That perspective also explains why project relevance cannot be inferred from semantic similarity to a call text. Real fit is a causal claim about contribution.

4.1 Horizon Europe: Relevance as a Causal Chain

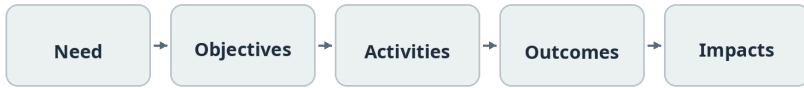
Horizon Europe is unusually explicit about the causal story it expects applicants to tell. For Research and Innovation Actions and Innovation Actions, the 2026-2027 General Annexes retain three award criteria: Excellence, Impact, and Quality and efficiency of implementation. The official application template goes further by asking applicants to explain target groups, expected outcomes, scale, significance, barriers, and measures intended to maximize uptake. This makes Horizon an excellent case study because relevance is not left as an undefined adjective; it is distributed across a chain of institutional questions.

The chain becomes clearer if we imagine a consortium proposing an AI tool for scientific-review automation. Excellence asks whether the objectives are clear and pertinent, whether the work goes beyond the state of the art, and whether the methodology is sound. Impact asks whether there is a credible pathway from the project's results to the expected outcomes and impacts described in the work programme. Implementation asks whether the plan, risks, resources, and consortium make that pathway executable. A fluent proposal can still fail if any of these links is weak. The criteria are separate because the institution is trying to prevent one strength from silently compensating for an unrelated weakness.

A useful practical discipline is to read every impact statement backward. Start with the claimed outcome and ask what actor would have to behave differently for that outcome to exist. Then ask what result would cause that behavior change, what evidence would make the result credible, what activity produces the evidence, and what assumption connects each step. If the chain breaks, the proposal has discovered an impact risk. This backward reading is harder to game than counting dissemination activities because it requires an explicit mechanism.

The value of the diagram lies in the arrows. Each transition makes a causal claim: the objectives address the need; activities can achieve

the objectives; outputs can be taken up; outcomes can contribute to longer-term impacts.



Every arrow is a causal assumption that needs evidence.

Figure 3. A simplified Horizon-style pathway from need to impact.

A convincing proposal supplies evidence for the vulnerable arrows rather than filling the boxes with ambitious nouns.

The 2026-2027 General Annexes preserve the familiar three award criteria for Research and Innovation Actions and Innovation Actions: Excellence, Impact, and Quality and efficiency of implementation. Under Impact, evaluators are asked about the credibility of pathways to expected outcomes and impacts and about likely scale and significance. This wording is more demanding than keyword alignment. It implies a chain in which proposed activities need mechanisms that plausibly connect them to outcomes outside the consortium.

Horizon Europe probably offers the most explicit contemporary form of institutional relevance assessment at scale. The 2026-2027 General Annexes retain, for Research and Innovation Actions and Innovation Actions, the triad of Excellence, Impact, and Quality and efficiency of implementation. Under Excellence, evaluators consider the clarity and pertinence of objectives, ambition beyond the state of the art, and the soundness of the methodology. Under Impact, the central question is the credibility of the pathways toward expected outcomes and impacts, followed by the quality of dissemination, exploitation, and communication measures. Version 5.1 of the application template, dated 22 January 2026, explicitly asks for the

project's unique contribution, target groups, scale and significance, external barriers, and mitigating measures.

Horizon proposals are often weakened by treating the call text as a vocabulary-matching exercise. The table reformulates familiar dimensions as questions about causal contribution rather than textual similarity.

Dimension	A stricter reading of relevance in a Horizon Europe proposal
Topic fit	Explicit mapping to scope and expected outcomes; necessary, but insufficient.
Problem relevance	Independent evidence that the need or gap exists and has meaningful magnitude.
Intervention relevance	An argument that the project's approach is better than, or complementarily necessary to, realistic alternatives.
Pathway credibility	The chain results → target groups → uptake → outcomes contains mechanisms, not merely aspirational verbs.
Scale / significance	Estimates include baselines, denominators, ranges, and assumptions rather than isolated percentages.
Barrier realism	Regulation, markets, user behavior, and external dependencies are treated as causal variables rather than footnotes.

Table 4. A stricter interpretation of relevance in a Horizon Europe proposal.

This reading makes proposal writing harder but evaluation more meaningful. The applicant must show not merely that the work belongs to the topic, but why the proposed mechanism is a credible part of the route from a European need to an expected outcome.

This design is more intelligent than the caricature "write an impact section." It tries to force a transition from rhetoric to mechanism. A result is not impact. A workshop is not uptake. A publication is not societal change. The project must explain who will use which result, why those actors would prefer it to alternatives, which barriers may prevent use, and what magnitude of effect is plausible.

Yet the same system creates its own pathologies. When a call defines destinations and expected outcomes in great detail, applicants learn to mirror the call's vocabulary. Proposals can become exercises in semantic alignment with policy rather than explorations of the best question. A rushed reviewer can confuse impeccable textual mapping with deep relevance. A consortium can generate dozens of KPIs without demonstrating the mechanism among them. Dissemination can become a list of channels, exploitation a vague promise, and impact a collection of European nouns.

The phrase 'credible pathway' deserves particular attention. A pathway is not a list of dissemination channels. It is a causal argument. If the project promises that a software platform will improve research productivity, the proposal should identify who will adopt it, what current process it replaces, what switching costs exist, which integrations are necessary, what evidence will convince users, and which external conditions might block adoption. If it promises policy influence, it should identify the decision window, the actors with authority, the form in which evidence must arrive, and the institutional mechanism through which change is possible.

Scale and significance are equally easy to confuse. Scale concerns how many people, organizations, systems, or markets could be affected. Significance concerns how important the effect would be. A tool used by ten laboratories to remove a catastrophic safety risk may have small scale but high significance. A communication campaign seen by a million people may have large scale and negligible effect. Horizon templates encourage applicants to quantify these ideas, but quantification is only useful when baselines, denominators, assumptions, and uncertainty are visible.

This perspective suggests a practical AI architecture for proposal support. The machine should not merely ask whether the words 'expected outcome' occur. It should build a traceable graph from call text to project objectives, work packages, deliverables, target groups, uptake mechanisms, outcomes, and impact claims. Every edge in that graph is a place where an evaluator can ask for evidence. The

resulting diagram is not decoration; it is a compressed causal model of why the project is relevant.

The same method improves KPIs. A KPI is valuable when it observes a causal link or decision-relevant state, not when it merely creates a large dashboard. Number of workshops is an activity measure. Number of organizations that integrate a tool is closer to uptake. Reduction in a validated error rate is an outcome measure. A proposal becomes more intelligible when its indicators are attached to distinct levels rather than presented as a single undifferentiated proof of impact.

4.2 ERC and EIC: Different Institutions, Different Relevance

European research funding becomes more intelligible when we stop assuming that every instrument means the same thing by relevance. The ERC is the clearest counterexample. For Starting, Consolidator, Advanced, and Synergy Grants in 2026, scientific excellence remains the sole evaluation criterion. The ERC explicitly focuses on the ground-breaking nature and ambition of the project together with the intellectual capacity and creativity of the Principal Investigator. It does not ask applicants to prove that their proposal fits a predefined policy topic or produces near-term societal impact.

The 2026 ERC proposal structure reinforces this philosophy. Part I, limited to five pages, is designed to make the case for the idea, the important question, the current state of knowledge, and the overall strategy. Feasibility is deliberately moved out of Step 1 and assessed in greater detail at Step 2 through Part II, which contains methodology, work plan, risks, and resource justification. This is more than a formatting change. It recognizes that an original idea can be prematurely killed if generalist reviewers demand implementation detail before they have understood why the question matters.

This institutional diversity also helps explain why 'impact' should not be used as a moral hierarchy among funding schemes. ERC's decision not to require near-term societal impact is itself a theory of

long-term relevance: society benefits by protecting space for investigator-driven questions whose applications may be unknowable. EIC's market and leverage criteria embody a different theory: public funding is justified when breakthrough innovation can be pushed across a financing or scaling gap. Collaborative Horizon topics embody a third theory: coordinated European action can accelerate defined transitions or capabilities. Each instrument is a different experiment in how to convert public resources into future options.

ERC makes the contrast especially clean in 2026: excellence remains the sole evaluation criterion for its main frontier-research grants, with panels focusing on the ground-breaking nature and ambition of the project and the intellectual capacity and creativity of the principal investigator. EIC Accelerator, by contrast, evaluates excellence, impact, and the level of risk, implementation, and need for Union support; its 2026 guidance asks about breakthrough character, market opportunity, team capability, and whether public support is needed to unlock the investment. The same word 'excellent' therefore sits inside different institutional objective functions.

Genuine European relevance should be separated into at least three layers. The first is topic fit: the proposal answers the topic as written. The second is problem relevance: the problem is real and sufficiently important independently of the existence of the call. The third is intervention relevance: the proposed project is a competitive intervention relative to real alternatives. Many evaluations are strong on the first layer and fragile on the third.

This is a place where AI can add immediate value. A good system can build a traceable matrix connecting call language, proposal claims, work packages, deliverables, KPIs, target groups, and expected outcomes. It can flag promises without mechanisms, KPIs without baselines, activities presented as impact, and scale claims without denominators. It can red-team the Impact section by asking what must be true for the pathway to work. It should not, however, decide on its own whether a problem is "European enough." That is a political and axiological choice, not a technical deduction.

Another advantage of auditable automation is that assessment can be compared with implementation. If relevance is scored at the beginning, we should return after three, five, or ten years and ask what actually happened. Otherwise, impact criteria remain an unverified rhetorical art. Horizon Europe has precisely the scale of data required for longitudinal validation: proposals, evaluations, funded projects, outputs, and programme-level monitoring. Research that connects these layers could materially change the way science is funded.

The European Innovation Council occupies a different part of the landscape. EIC Accelerator evaluation examines excellence, impact, and the level of risk, implementation, and need for Union support. The 2026 criteria ask about breakthrough nature, technology readiness, intellectual property, market opportunity, team capability, risk, implementation, and investment leverage. Here relevance includes a credible relationship to a market and to European strategic value. An idea can be scientifically fascinating yet irrelevant to the Accelerator because it is too early, lacks a plausible commercialization path, or does not need the instrument's form of support.

This comparison is conceptually important. A project rejected by one instrument is not therefore an irrelevant project. It may be relevant to a different decision regime. ERC asks whether the research is excellent and ground-breaking. A collaborative Horizon topic asks whether a consortium can contribute to specified expected outcomes. EIC asks whether innovation can become a scalable venture under conditions in which public intervention is justified. Relevance is institutional before it becomes numerical.

For AI-based review, the implication is strict: the context object must encode the funding instrument before any scoring begins. A model trained to reward Horizon-style impact language could distort ERC evaluation by penalizing fundamental research that intentionally lacks a short path to adoption. A model optimized for venture-market evidence could distort a Research and Innovation Action by underweighting public goods. A trustworthy system must

know not only what a proposal says, but which evaluative world it has entered.

4.3 From Call Fit to Portfolio Relevance

The most common failure in European proposal writing is to confuse topic fit with relevance. Topic fit asks whether the proposal answers the scope and expected outcomes stated by the call. It is necessary because the Commission is allocating money for a defined purpose. But fit is only the outer gate. Problem relevance asks whether the underlying need is real and important independently of the wording of the call. Intervention relevance asks whether this particular project is a plausible and competitive way to address that need. Portfolio relevance asks whether funding the project adds something useful to the set of projects already selected or likely to be selected.

This four-layer view explains why semantic matching can become dangerous. Generative AI can produce proposal text that mirrors every phrase in a work programme. A keyword or embedding system may therefore report extraordinary alignment. Yet the same project may rely on an obsolete technical assumption, duplicate existing funded work, or offer a pathway whose target users have no incentive to adopt the result. Textual fit is the easiest layer to automate and the easiest to game. The harder work begins when the evaluator asks whether the intervention changes the counterfactual.

European-project relevance also has a governance dimension. A topic does not describe only a scientific problem; it is the product of negotiation among European strategies, sectoral policies, Member States, research communities, and industry. "Fit with the topic" is therefore relevance to an institutional priority that has already been selected. That relevance is legitimate for allocating a programme's budget, but it should not be confused with a universal judgment about the importance of the underlying problem. A project can be scientifically profound and irrelevant to a particular funding instrument; another can answer the topic perfectly while remaining incremental relative to the global frontier.

EIC Pathfinder Challenges make portfolio logic unusually explicit: after individual evaluation, proposals can also be considered for their contribution to a coherent portfolio, including diversity and synergies among projects. This matters for relevance theory because a proposal can be valuable partly through complementarity. Two individually strong projects may be redundant together, while a slightly less conventional project may complete a capability that makes the whole portfolio more useful.

The portfolio view also turns evaluation back onto the call itself. If successive rounds attract proposals that are formally compliant but repeatedly fail to produce the intended outcomes, the defect may lie in programme design rather than applicant quality. The work programme may be asking for the wrong intervention, specifying an implausible timescale, fragmenting a system problem into incompatible projects, or rewarding easily measurable outputs over harder institutional change. A mature relevance infrastructure should therefore evaluate calls and programmes with the same seriousness used to evaluate proposals.

Europe has an unusual advantage here because the evidence can be longitudinal and comparative. Topic text, evaluations, funded portfolios, deliverables, patents, standards, policy uptake, follow-on investment, and later programme revisions can be linked into a history of expectations and consequences. Used carefully, such a history could reveal which kinds of claimed pathways were predictive, which barriers were routinely underestimated, and where portfolio diversity created resilience. Used badly, the same archive could train an automated reviewer to reproduce yesterday's preferences. Programme learning must therefore preserve counterexamples and failed predictions rather than only the narratives of successful projects.

A second tension concerns specific needs, expected results, and expected outcomes. The Horizon template is useful because it forces a project to show the path between them, but that path is not merely a diagram. For every major result, there should be an actor capable of adopting it, an incentive to adopt it, compatibility with real infrastructure, and an account of barriers. If a consortium promises

a standard, it should show who participates in standardization. If it promises industrial adoption, it should show integration cost and purchasing authority. If it promises policy change, it should identify the institutional window and channel of influence. European relevance thereby becomes mechanism engineering rather than vocabulary compliance.

A third tension is scale versus significance. An intervention may affect ten organizations in a transformative way or ten million people in an almost imperceptible way. Aggregating both into a single estimate produces false comparability. A strong proposal should separately describe how many actors can be reached, how important the change is for each category, and how much of that effect can plausibly be attributed to the project. The same principle will matter for AI-generated art and content: reach is not the same as intensity of experience, nor is it the same as cultural persistence.

Finally, there is the Commission's portfolio relevance. A call should not be optimized merely to select the proposals that most closely resemble the topic text. If all funded projects use the same architecture, the same class of partners, and the same hypothesis about the market, the programme becomes fragile. A mature process could add, after individual scoring, an explicit portfolio-construction stage: coverage of different hypotheses, risk distribution, complementarity of results, and avoidance of redundancy. This is a surprisingly fertile area for AI because models can inspect hundreds of proposals simultaneously and detect overlaps that isolated reviewers do not see.

Portfolio relevance is particularly underdeveloped. Calls usually evaluate proposals one by one and then rank them, although programmes ultimately care about the combined effects of the funded set. Suppose the top ten proposals all use the same model architecture, require the same scarce dataset, and assume the same regulatory future. Individually they may be strong; collectively they create systemic risk. A more resilient portfolio might fund several mainstream projects, a smaller number of high-risk alternatives, shared infrastructure, and one or two approaches based on a radically different technical premise.

AI is well suited to this portfolio layer because it can compare across the entire call. It can cluster approaches, detect repeated dependencies, map which expected outcomes are oversupplied or neglected, and simulate how a failure in one shared assumption would affect the portfolio. The model should not choose the political balance on its own. It can make the balance visible. This is a recurring pattern in the book: automation is most defensible when it expands the evidence available to accountable decision-makers rather than replacing the value judgment.

Europe also offers the possibility of something rarer: longitudinal validation. The ecosystem already contains structured proposal data, evaluation records, funded projects, deliverables, publications, patents, software, policy interactions, and programme monitoring. Connecting those layers while respecting confidentiality would allow researchers to ask whether *ex ante* relevance judgments actually predict *ex post* outcomes, whether particular criteria select conservative portfolios, and which impact claims were merely well written. A funding system that learns from its own past decisions could turn evaluation from administrative ritual into an empirical science.

European programmes demonstrate that relevance is policy encoded in an evaluation procedure. Horizon, ERC, and EIC do not disagree because one understands quality and the others do not; they allocate scarce resources toward different public purposes and therefore need different evidence.

With that institutional lesson established, the book can now leave science without losing conceptual discipline. Literature is the first test of how far the grammar travels.

5. Literature: Relevance Beyond Preference

Imagine a publisher that receives one million competent novels every week. The old question - can this person write? - becomes almost irrelevant because fluency is abundant. The publisher needs to know which works deserve scarce editorial attention, which ones add something to the catalog, which voices would otherwise disappear, which books may matter to a particular community, and which apparently excellent texts are simply high-quality repetitions of patterns already available in enormous quantities.

Literature is a useful stress test because relevance here cannot be reduced to factual correctness or utility. A novel can matter through style, perspective, emotional structure, moral imagination, formal experiment, cultural testimony, or the later work it makes possible. Some of these dimensions can be measured indirectly; none should be confused with simple preference prediction.

AI makes this more than an aesthetic debate. If synthetic text becomes a major share of what can be published, the selection regime will shape the future distribution of language and ideas. The chapter therefore moves from the relevance of one book to the relevance of a whole literary ecosystem.

5.1 Craft Is Only the Entrance Fee

Literature is an unforgiving test for relevance because it exposes the inadequacy of utility and preference. A novel can matter without solving a problem, teaching a fact, or satisfying the reader in any simple way. It may disturb, confuse, delay understanding, or create a vocabulary for an experience that readers did not previously know how to name. Its relevance can emerge from the relation between form, language, historical moment, and reader rather than from a measurable external objective.

Creativity research nevertheless offers useful structure. The Consensual Assessment Technique, associated with Teresa Amabile and later work, asks knowledgeable judges to rate creative products without requiring a complete formal definition of creativity. The Creative Product Semantic Scale separates aspects such as novelty, resolution, and elaboration. These approaches matter because they recognize multidimensional judgment and inter-rater variation instead of treating creativity as a single latent number that a model can simply recover.

Literature is a difficult test for any theory of relevance because it resists reduction to utility. A story does not need to solve an external problem in order to matter. It can create a problem, name an experience that previously lacked a vocabulary, alter how a reader understands a relationship, or introduce a form that later writers will exploit. Literary relevance is generative in an important sense: it does not merely fit a context; it can change the context.

The empirical literature on computational creativity should also make us cautious about apparent objectivity. LLM evaluators can show high agreement on surface dimensions such as fluency or elaboration while remaining weaker on culturally specific and context-dependent aspects of creativity. Agreement is therefore evidence about consistency, not proof that the criterion captures literary importance.

The psychology of creativity nevertheless offers useful tools. The Creative Product Semantic Scale decomposes evaluation into novelty,

resolution, and elaboration/synthesis. The Consensual Assessment Technique proposes that a group of competent judges can provide a valid measure of creativity without a completely formal definition. These methods matter precisely because they separate creativity from simple correctness. Recent work on creative text uses rubrics inspired by Torrance, human evaluation, and LLM-as-a-judge methods for originality, flexibility, elaboration, and related dimensions.

The results are mixed, which is healthier than a simple verdict on "LLM creativity." A CHI 2024 study comparing LLM stories with New Yorker-level writing, while controlling for factors including length and plot, concluded that models remained far from top human writers on the creative dimensions examined. Other studies find performance close to ordinary human writers on shorter tasks or show that models can produce many variants, increasing collective creativity. The distinction between "a good average output" and "a memorable work" remains essential.

For relevance, dimensions missing from standard creativity benchmarks should be added. The first is contextual resonance: does the text enter a real conversation or merely recycle stylistic signals? The second is density of consequence: how many interpretations, connections, and questions does it generate without collapsing into vagueness? The third is cultural novelty: does it introduce a combination of ideas that changes what seems possible within a genre? The fourth is persistence: what remains after the immediate effect of fluent prose has faded?

Recent work on large language models reinforces this caution. Studies of AI creative writing produce mixed results depending on task length, comparison group, rubric, and evaluator. Some models can outperform average participants on constrained divergent-thinking tasks or produce fluent short-form writing. Other work, including CHI 2024 comparisons with professional literary writing, finds a substantial gap when the benchmark involves higher-level originality and craft. The disagreement is informative: 'creativity' changes meaning when we move from generating many

plausible variations to producing a work that remains memorable after close reading.

For relevance, literary evaluation needs dimensions that creativity benchmarks often miss. Contextual resonance asks whether the work enters a real conversation rather than merely imitating the stylistic markers of importance. Interpretive density asks whether repeated reading reveals productive connections rather than arbitrary ambiguity. Durability asks whether the work depends entirely on a passing trend or has structures that can survive changes in context. Generativity asks whether the work creates questions, metaphors, or forms that other people can continue.

None of these dimensions can be read directly from first-impression liking. A highly fluent model may be particularly good at producing texts that feel complete and familiar on first encounter. Great literature is often less cooperative. It may leave gaps, resist paraphrase, or violate a reader's current expectations. An evaluator designed only to maximize immediate preference would select a smooth literary present at the cost of a more surprising literary future.

5.2 Editorial Selection When Novels Are Infinite

Generative abundance changes the editorial problem more radically than the writing problem. If a system can produce one acceptable novel after another, producing another novel has low marginal informational value. The scarce task becomes discovering which work deserves human time. That means future editorial systems will increasingly resemble research-review systems: they will compare a text with a corpus, test claims of originality, inspect long-range coherence, solicit multiple expert perspectives, model candidate audiences, and preserve uncertainty where evaluation is unstable.

A literary relevance dossier could begin with craft as a threshold rather than a final score. It would test whether characters, voice, structure, and temporal logic remain coherent across the full work. A separate originality component would retrieve close semantic,

narrative, and stylistic neighbors to distinguish substantive difference from superficial recombination. Another component would model likely interpretive communities, not merely demographic market segments. A historical critic could ask how the work relates to traditions and whether those relations are conscious, derivative, subversive, or unexpectedly productive.

These dimensions cannot be assessed only through instantaneous preference. Language models tend to reward local coherence, explicit formulation, and completeness - exactly the qualities that can make a text competent and forgettable. Relevant literature tolerates ambiguity, dissonance, gaps, and interpretive risk. A good AI evaluator should be capable of saying "I do not know yet" and of modeling disagreement among reader communities.

The problem can be stated as a distributional one. If every selected novel is individually polished but all selected novels are close to one another in theme, voice, structure, or worldview, the editorial system has failed at the level of the catalog. A relevance system for abundant literature should therefore optimize both local quality and marginal contribution to collective diversity.

In a world with millions of generated novels, the editorial problem will increasingly resemble the research problem. We will need to examine novelty claims, relationships with the corpus, long-range coherence, possible audiences, meaning, and option value. Literature nevertheless preserves a crucial difference. In science, evidence can invalidate a claim; in literature, a new interpretation can create relevance without invalidating older interpretations. The selection system must know which epistemic world it is operating in.

Literary evaluation becomes more defensible when the system records several dimensions separately instead of hiding them inside a synthetic 'quality' score. The table gives a compact example of the evidence a dossier might preserve.

Dimension	A relevance dossier for AI-generated literature
Craft	Coherence, stylistic control, structure, and appropriateness of execution; necessary but not sufficient.
Originality	Semantic and formal distance from the corpus, evaluated carefully for pastiche and superficial recombination.
Resonance	Relation to themes, experiences, or tensions recognizable to an audience without reducing value to popularity.
Generativity	Capacity of the text to produce interpretations, questions, and further intellectual continuations.
Memorability	Persistence of images, ideas, structures, or voices after immediate consumption.
Plurality	Differences among evaluators and communities preserved as information rather than erased by a premature average.

Table 5. A relevance dossier for AI-era literature.

None of the rows settles whether a novel deserves publication. Their purpose is to make editorial disagreement legible and to prevent preference prediction from masquerading as cultural judgment.

Literature also has a property that complicates automated evaluation: relevance is often retrospective and intertextual. A book can appear conventional when it is published and become relevant when a new social context reinterprets it. Conversely, a novel perfectly synchronized with a momentary fashion may disappear when its references stop functioning. An evaluator therefore needs to distinguish topicality from durability. Topicality measures fit with the present conversation; durability concerns the capacity to generate readings after the immediate context has changed. The latter cannot be fully observed *ex ante*, but indicators can be preserved: interpretive density, dependence on trends, and the variety of communities that find meaning in the text.

This leads to the problem of the canon. Expert consensus is not a neutral oracle. Critics, publishers, universities, and prizes help construct visibility. If an evaluator is trained on canonical texts and

on the judgments of institutions that consecrated them, the model may learn both quality and the history of power. A literary relevance dossier should separate similarity to the canon from arguments for value, include evaluators from different traditions, and flag cases in which the verdict depends strongly on cultural familiarity.

For AI generation, selection becomes more important than individual scoring. If a model produces one hundred acceptable novels, a platform should not simply choose the first ten by "quality score." It should build a portfolio with different voices, structures, levels of risk, and audiences. In this sense, the editorial practice of the future resembles portfolio science: retain some projects with a high probability of success, some unusual explorations, and some works that optimize no existing audience but may create one.

Human plurality is essential because literary disagreement is not simply measurement noise. Different communities can legitimately find different relevance in the same work. A migration narrative may carry meanings unavailable to a reader who lacks that social experience. Experimental form may be legible to critics steeped in a tradition and opaque to a mass audience. Instead of forcing those views into one average, an AI-supported system can preserve the structure of disagreement and show which claims are community-dependent.

The canon creates an additional bias problem. Models trained on canonical literature learn both artistic achievements and the institutional history that made some works visible while others disappeared. Similarity to the canon can therefore encode power alongside quality. A responsible evaluator should explicitly separate 'resembles historically consecrated work' from 'has a defensible argument for relevance.' It should also test how verdicts change when author identity, prestige, nationality, or publication context are hidden or altered.

Finally, selection should occur at portfolio level. A publisher or platform choosing fifty works should not simply take the top fifty by average score. It might deliberately preserve different voices, formal risks, audience scales, and degrees of uncertainty. Some books

should be safe bets; some should serve small communities deeply; some should explore forms for which the system has little training data. In this sense the future editor becomes a portfolio scientist, managing not only expected quality but cultural option value.

5.3 Relevance Across Time and Communities

Literary relevance cannot be inferred from the first audience alone. A novel enters several overlapping systems: the experience of individual readers, a genre tradition, a publishing market, educational institutions, criticism, libraries, translation networks, and eventually cultural memory. Each system notices different properties and operates on a different clock. Immediate readability may matter to a commercial publisher; formal innovation may matter to critics; social testimony may matter to a community; unusual language may matter only after later writers reveal what it made possible. A single launch-period rating collapses these distinct processes into one noisy moment.

This temporal structure is familiar from science. The value of a mathematical technique may remain latent until another field needs it. Literature has analogous delayed pathways. A work can supply a form, voice, image, or moral vocabulary that later creators and communities reuse. The relevant evidence is not only sales or prizes but quotation, adaptation, imitation, translation, teaching, critical reinterpretation, and the emergence of new works that could not easily have been written in the same way without it. None of these signals is pure, but together they provide a richer history of uptake.

Communities also differ in what they can see. A work that appears marginal to a dominant audience may be central to readers for whom it articulates an experience rarely represented elsewhere. Conversely, a text that perfectly reflects the assumptions of a dominant group may receive high immediate fluency ratings while adding little interpretive range. This is one reason diversity cannot be reduced to demographic bookkeeping. The evaluative question is whether a cultural system retains enough heterogeneity of

perspective and form to discover value that its dominant model does not already predict.

Generative AI intensifies the issue because it can personalize prose toward increasingly narrow audiences. In the benign version, this means accessibility: a story can be adapted to age, language, reading ability, or cultural context. In the dangerous version, literature becomes a frictionless service optimized to confirm the reader's current preferences. A recommendation engine can then select from generated texts that were themselves optimized by related preference models. The whole pipeline may become extremely good at satisfaction and extremely poor at surprise.

The Science Advances experiment by Doshi and Hauser offers a concrete warning. Writers given generative-AI ideas produced short stories that evaluators rated as more creative and enjoyable, especially when the writers were initially less creative. Yet the AI-assisted stories were more similar to one another. The individual artifact improved while the collective distribution narrowed. This is a crucial distinction for relevance systems: maximizing the average score of each item is not equivalent to maximizing the long-run value of the corpus.

A literary evaluator should therefore report both artifact-level and population-level signals. At the artifact level, it can examine coherence, craft, originality relative to a corpus, emotional architecture, semantic density, and the strength of competing interpretations. At the population level, it should ask what the work adds to the available distribution: Does it introduce a rare structure, perspective, vocabulary, or combination? Does it merely occupy an already crowded cluster? Does it make the cultural portfolio more resilient to homogenization? These questions are more difficult than predicting likes, but they are closer to what becomes scarce under abundance.

Authorship complicates evaluation further. Readers may legitimately care whether a text records a lived experience, documents human testimony, or represents a technical feat of machine generation. Source information can therefore be relevant

rather than a bias to be eliminated. The problem begins when source labels silently contaminate a score that is presented as if it measured only textual properties. A rigorous system should separate judgments about the artifact from judgments about provenance, labour, authenticity, and institutional meaning.

This separation also prevents a common mistake in debates about AI literature. Asking whether an AI can 'really be creative' can become metaphysical before it becomes useful. For selection, the operational questions are more precise: What properties does the text exhibit? How different is it from relevant precedents? What effects does it have on readers? Which interpretations survive informed critique? What role does the production process play in the work's meaning? Which kinds of value would disappear if provenance were hidden? These questions can be studied without pretending that one philosophical definition settles the entire domain.

Long-term cultural relevance therefore resembles a portfolio with memory. Libraries, archives, publishers, and digital platforms should not only rank what appears best now; they should preserve enough variety for future re-evaluation. The cost of storage for digital text is tiny compared with the cost of attention, so preservation and promotion can be separated. A work need not occupy today's scarce recommendation slot to deserve survival in tomorrow's searchable cultural memory.

Preservation itself is not neutral. What receives stable identifiers, metadata, translations, searchable text, critical editions, and institutional care becomes easier for future readers and models to rediscover. What survives only as an obscure file may be technically present yet practically absent. A relevance system concerned with long horizons should therefore treat discoverability infrastructure as part of cultural memory. This is analogous to research data stewardship: keeping an artifact is not enough if its provenance, context, and relationships are lost.

The same argument changes how we should think about canons. A canon can be useful as compressed cultural memory, giving

communities a shared set of references, but it becomes dangerous when compression is mistaken for completeness. Under generative abundance, the sensible response is neither to abolish canons nor to freeze them. It is to maintain a visible core together with mechanisms for revision, challenge, and rediscovery. Relevance then becomes partly a governance question: who may propose additions, what evidence can reopen an old judgment, and how much attention should be reserved for works that established evaluators initially fail to understand?

That distinction between preservation and promotion will recur in film and visual art. It is one of the most useful ideas research assessment contributes to culture: not every uncertain object deserves equal present resources, but uncertainty is not a license to erase the future.

Literary relevance is therefore both local and ecological. A work can be excellent for one reader while adding little to the collective repertoire, or initially difficult while opening a new space that later readers learn to inhabit. AI makes the ecology impossible to ignore because it can improve individual outputs while narrowing aggregate diversity.

Film adds industrial scale and algorithmic recommendation, allowing us to examine what happens when relevance is not only judged but actively mediated by platforms.

6. Film: Relevance Beyond Recommendation

Film sits between art, industry, technology, and mass recommendation. It is expensive enough that project selection has historically occurred before production, yet digital distribution and synthetic media are moving more experimentation downstream. At the same time, streaming platforms already mediate attention through recommender systems whose objectives are often behavioural rather than cultural.

A film therefore offers several distinct objects of relevance at once: a production project, a finished artistic work, a commercial asset, a personal recommendation, a social reference point, and an archival object. Many arguments about algorithmic culture become confused because they evaluate one of these objects and assume the conclusion applies to all of them.

The chapter separates those layers and asks what happens when AI can generate not only recommendations but scripts, scenes, edits, localized versions, and eventually whole variants. The core issue becomes whether personalization broadens access to culture or quietly replaces shared cultural discovery with optimized individual consumption.

6.1 Four Films Hiding in One Film

Film contains several relevance problems inside one artifact. The viewer asks, 'Is this worth my evening?' The studio asks whether the film can recover investment and strengthen a slate. The critic asks what it does formally and historically. A cultural institution asks what it contributes to representation, heritage, shared memory, or access. These questions overlap but cannot be reduced to one ranking without losing information.

Recommender-systems research already discovered a version of this problem. Accuracy alone can create repetitive recommendation lists that perfectly mirror past behavior. The field therefore studies diversity, novelty, serendipity, coverage, and fairness as beyond-accuracy objectives. Serendipity is especially relevant because it combines surprise with usefulness: a random film the viewer dislikes is not serendipitous, while an unexpected film that expands taste may be. This is a direct analogue of exploration in research funding.

Film makes it obvious why relevance must be assessed at several levels. For a streaming platform, a film is relevant if the user will choose and appreciate it. For a studio, it may be relevant if it produces revenue, retention, or reputation. For a critic, relevance may mean formal innovation or position within a tradition. For cultural history, the question becomes whether the film changes visual language, the representation of a community, or the memory of an era.

This is not merely a philosophical distinction. Producers, platforms, critics, public broadcasters, film archives, educators, and viewers are all legitimate evaluators, but they have different purposes. The same film can be a weak commercial proposition, a strong artistic achievement, an important local document, and an ineffective personal recommendation. The evaluation should identify which claim is being made before deciding what evidence counts.

Recommender systems have already learned that accuracy is not enough. A list perfectly fitted to a user's history can become an elegant prison. Hence the importance of novelty, diversity, and serendipity: a good recommendation should not merely confirm taste, but sometimes expand it. The literature on serendipity defines value precisely as a combination of relevance and surprise. An unexpected but fitting film may be more valuable than the tenth film almost identical to those already watched.

The same film can succeed under one evaluative frame and fail under another. The table names the layers that are most often mixed together when discussing recommendation, success, and cultural value.

Layer	A distinct meaning of film relevance
Personal	Fit with the state, interest, and context of a particular viewer.
Market	Capacity to generate audience, revenue, retention, or commercial effect.
Critical	Formal, thematic, and historical contribution as seen by competent evaluators.
Cultural	Effect on collective language, representation, memory, and the longer-term canon.
Tension	A film can be weak on one layer and strong on another; averaging them can destroy the information that matters.

Table 6. Distinct meanings of film relevance.

Once the layer is explicit, evidence can be chosen appropriately. Viewer completion is useful for personal engagement; it is weak evidence for long-term cultural memory. Festival recognition can indicate critical attention; it is not a direct measure of commercial viability.

This observation is crucial for AI film generation. If production cost falls radically, platforms will have a powerful incentive to generate hyper-personalized content. But strict optimization for click-through, watch time, or rating can produce a culture in which everyone receives what the algorithm thinks they already want.

Maximum personalized relevance can destroy shared cultural relevance.

The BFI and other cultural-impact programmes try to capture effects that exceed box office. Research on canon formation shows that cultural memory is produced by critics, institutions, lists, archives, and repetition. Some films are forgotten not because they lack value but because they never enter the machinery of consecration. An AI evaluator trained on the existing canon risks confusing historical consecration with intrinsic merit.

Market relevance is measurable but still multidimensional. Box office, subscriptions, completion rate, retention, licensing revenue, and audience growth answer different business questions. A film can be commercially weak on release and later become valuable through catalog persistence. A documentary can have a small audience but large significance for a policy community. A franchise entry can generate enormous revenue while contributing little to the evolution of film language. Economic data are therefore evidence for one layer, not a universal verdict.

Critical and cultural relevance operate on longer and more institutional timescales. Festivals, reviews, archives, restorations, film schools, essays, remakes, visual quotations, fan communities, and references in later works all contribute to canon formation. These mechanisms reveal influence but also create it. A film unavailable for decades cannot generate the same traces as one continuously distributed. Absence of cultural evidence may therefore reflect distribution history as much as intrinsic insignificance.

For AI-generated film, the temptation will be to optimize the easiest layer: predicted personal satisfaction. That layer is valuable, but the larger ecology requires separate models for market, critical, and cultural relevance. Otherwise the system will confuse 'I am likely to watch this tonight' with 'this deserves to occupy cultural attention.'

6.2 When Personalization Erodes Shared Culture

Hyper-personalized synthetic video makes the conflict between personal and systemic relevance vivid. Imagine that a platform can generate a feature film for each viewer, tuned to preferred pacing, actors, themes, humor, and emotional arc. At the individual level this could be extraordinarily relevant. At the social level it could erode the shared reference points that make public conversation possible. A culture in which everyone receives a perfect private movie may be less culturally coherent than one in which imperfect common works are discussed together.

Shared relevance is partly a public good. People derive value not only from the artifact but from knowing that others have encountered it. A film becomes a shorthand in conversation, journalism, education, and politics. This network effect cannot be captured by summing individual preference scores. It suggests that recommendation portfolios should sometimes maximize common exposure or bridge communities rather than only optimize isolated users.

For film, relevance should be separated into four layers that should not be automatically aggregated: personal relevance, market relevance, critical relevance, and cultural relevance. A mature generative system should be able to optimize these layers differently and, at the portfolio level, preserve content that explores small audiences but has high potential for cultural diversification.

The strongest version of the concern is not that algorithms always narrow consumption. Evidence is mixed and depends on design. The deeper concern is institutional: when one optimization layer mediates discovery for a large population, the values encoded in that layer acquire cultural power. Even a diversity-aware recommender should therefore make its diversity objective explicit: genre variety, creator variety, viewpoint variety, temporal variety, or movement across embedding clusters are not interchangeable.

Here the research lesson returns. A healthy portfolio does not fund only projects with the highest probability of average success. It also preserves high-risk, high-uncertainty bets. Culture needs the same mechanism. Relevance is not only match; it is also the system's capacity to create encounters the user did not know to request.

Film adds another complication: it is simultaneously an artistic object, an economic product, and a social infrastructure of representation. A recommendation system usually optimizes the viewer's present decision; a studio optimizes a portfolio and investment recovery; a cultural institution may pursue diversity, heritage, or access. These objectives can be legitimately incompatible. "Relevant film" has no meaning without the role of the evaluator. An agent recommending something for an evening and a public film fund should not share the same utility function.

Cultural relevance can leave traces richer than box office: visual citations in other works, use in education, appearance in public discourse, remakes, festivals, restorations, fan communities, and shifts in representation. Yet these signals are themselves produced by institutions. A restored film gains visibility; an inaccessible one cannot generate traces. AI can help reconstruct networks of influence, but it must distinguish absence of evidence from evidence of absence.

When the cost of video production falls dramatically, a new problem of attention ecology appears. It is not enough for every film to be reasonably good. An ecosystem with millions of personalized productions can reduce the shared experiences that make cultural conversation possible. System-level relevance may deliberately require shared reference points rather than maximum personalization. The metric of individual recommendation and the health of culture can pull in opposite directions.

There is also a distributional question. Personalization systems trained on existing engagement can allocate most visibility to already dominant genres and identities while relegating smaller communities to narrow niches. Fairness metrics help, but cultural evaluation requires more than demographic parity. A system may

need to preserve works that expose audiences to unfamiliar perspectives, support minority-language production, or maintain archives that have low immediate commercial value.

The research-funding analogy is useful here. Frontier funders preserve high-risk projects because exploitation of known winners alone eventually stagnates. Cultural systems need exploration budgets for the same reason. Some fraction of recommendation and production capacity should be reserved for uncertain works with high novelty, bridge potential, or cultural option value. The exact fraction is a policy choice, but making it zero is also a policy choice - one that silently optimizes the present distribution of taste.

An AI relevance system for film should therefore report several maps at once: predicted personal fit, market potential, critical/formal contribution, cultural positioning, and portfolio effects. The most interesting decisions will be those where the maps disagree. That disagreement is not a bug. It is where institutions reveal what they are actually trying to preserve.

6.3 Shared Culture, Diversity, and the Long Tail

Film makes the difference between personal relevance and system relevance unusually visible. A streaming platform can be excellent at finding the film most likely to hold one viewer's attention tonight while simultaneously weakening the circulation of works that sustain a common cultural vocabulary. The two outcomes are not logically opposed, but they are not the same objective. Recommender systems mostly optimize observable behaviour at the individual level; cultural institutions often care about diversity, exposure, memory, and shared reference points at the population level.

The popular language of 'filter bubbles' can oversimplify this tension. A systematic review by Areeb and colleagues found evidence of filter-bubble effects but also substantial dependence on algorithm design, user behaviour, and how diversity is measured. Recent work on music consumption similarly suggests that algorithmic curation can sometimes broaden rather than narrow what people consume. The responsible conclusion is not that

recommendation inevitably fragments culture, nor that the concern is imaginary. It is that diversity has several scales and needs to be measured directly rather than inferred from a slogan.

For film, at least four kinds of diversity matter. Catalog diversity asks how varied the available works are. Exposure diversity asks what the system actually shows to users. Consumption diversity asks what people choose and finish. Interpretive diversity asks whether audiences encounter different aesthetic languages, social perspectives, production traditions, and historical periods rather than merely different titles. A platform can score well on the first three and still be narrow on the fourth if most recommendations occupy the same underlying narrative and stylistic space.

The economics of streaming add another layer. Recommendation is not a neutral discovery service; it sits inside a business model. A system may prefer content that reduces churn, supports expensive proprietary franchises, or can be marketed across many regions. Those objectives may be legitimate for a company, but they should not be mislabeled as cultural relevance. The more AI participates in commissioning and generating films, the stronger the feedback loop becomes: observed engagement shapes recommendations, recommendations shape demand, demand shapes what gets generated, and the generated catalog then becomes the evidence on which future models are trained.

This is why the long tail matters differently in an AI era. Digital distribution made it technically possible to store and offer far more works than physical cinemas or video stores could. Generative media increases the tail by orders of magnitude. Yet attention does not scale with storage. The practical bottleneck moves to interfaces, critics, search, recommendation, and social coordination. A vast catalog without effective plural curation is not cultural abundance in the meaningful sense; it is uninspected storage.

A relevance-aware film platform would therefore expose trade-offs rather than hide them. One recommendation lane could optimize immediate personal fit. Another could deliberately maximize discovery outside the user's established cluster. A third

could surface historically important or locally significant works. A fourth could preserve a shared civic or educational canon while remaining open to revision. The point is not to force everyone through the same curriculum. It is to recognize that a culture can have legitimate objectives that are not reducible to individual click probability.

Evaluation should also distinguish the film as a finished work from the film as an industrial project. Before production, relevance concerns audience hypotheses, artistic ambition, financing, technical feasibility, rights, distribution, and cultural position. After release, new evidence appears: completion rates, reviews, box office, festival selection, quotation, remakes, classroom use, community response, and long-term rediscovery. A system that uses only commercial outcomes to train future selection will learn to imitate the market it helped create.

Synthetic media creates a further problem: versions may proliferate. A story can be re-edited for different languages, lengths, age groups, actors, or even individual viewers. At that point the unit of evaluation is no longer a single immutable film. It becomes a family of artifacts generated from a production model. Relevance assessment must decide which properties belong to the underlying work and which belong to a particular rendering or audience adaptation.

The mature response is not nostalgia for scarcity. Traditional film industries excluded enormous amounts of talent and routinely made expensive failures. AI can lower barriers, expand experimentation, improve accessibility, and make niche stories economically possible. The important question is which selection institutions replace the old bottlenecks. If the answer is only engagement prediction, cultural production will have traded one narrow gatekeeper for another.

Film therefore teaches a general lesson about relevance at scale: recommendation, preservation, commissioning, and cultural memory are distinct functions. A single model should not silently perform all four.

Public-service media provide a useful institutional counterexample to pure engagement optimization. Their mandates typically include access, education, representation, cultural preservation, and service to audiences that may be commercially unattractive. An AI curation system operating under such a mandate would need to justify why a recommendation contributes to several public purposes rather than simply predicting watch time. This is close to multi-criteria research funding: the objective function is explicitly plural, and disagreement about weights is a legitimate policy question rather than a model error.

The same principle applies to commissioning. If generative systems can cheaply produce trailers, scripts, storyboards, or synthetic pilots, a platform can test audience response before committing to a full production. That can reduce waste, but it can also create a selection bias toward ideas whose value is visible in short tests. Slow films, unfamiliar narrative forms, or works whose effect depends on sustained attention may look weak in the experimental proxy. The test regime can therefore shape the art that survives it, just as grant criteria shape the science that gets proposed.

A mature film relevance system should keep at least two memories: a memory of audience response and a memory of its own selection errors. Works initially predicted to fail but later rediscovered are especially valuable training cases because they reveal which forms of relevance the earlier model could not see. Without such longitudinal error analysis, recommender systems can become increasingly confident in a cultural world partly produced by their own previous choices.

Shared culture also has a coordination value that individual preference models rarely observe. People use films as reference points in conversation, education, political argument, humor, and collective memory. A work can therefore matter because many differently situated people have encountered it, even when it is nobody's perfectly optimized personal choice. This does not justify imposing a single canon; it explains why some institutions rationally spend attention on common exposure rather than maximizing individualized satisfaction.

A practical system can protect this value without returning to scarcity-era gatekeeping. It can reserve an explicit exploration budget for unfamiliar works, maintain multiple independent curatorial models, and record why an item was surfaced: personal fit, historical importance, local significance, formal novelty, or contribution to portfolio diversity. For synthetic media, lineage should also be preserved across versions so that cultural response to one adaptation is not silently treated as evidence about every descendant. The goal is not less personalization, but personalization embedded inside a wider ecology of discovery.

The lesson from film is that discovery is itself an institution. A recommendation system determines which evidence of relevance can ever be generated because unshown works cannot be chosen, discussed, or remembered. Evaluation and exposure form a feedback loop.

Design turns that abstract problem back toward the physical and technical world, where claims can often be challenged by tests and failure becomes informative rather than merely unpopular.

7. Design and Technical Artifacts: Relevance Under Constraints

Design looks easier than literature because constraints can be written down. A bridge has loads, a medical device has safety requirements, an interface has tasks, a protocol has invariants, and software has tests. Yet every experienced engineer knows that passing the specified tests is not enough. Real systems fail at interfaces, under maintenance, through misunderstood users, through dependencies, through adversaries, and through assumptions that were never included in the benchmark.

This makes design the ideal domain for connecting relevance to evidence. A design is a bundle of claims about use under constraints. AI can generate many variants, but each variant is only as meaningful as the tests that distinguish it from the alternatives. The cost of generation falls; the value of experimental design rises.

The chapter begins with physical and interaction design, then extends the argument to software and research infrastructure, where auditability, interoperability, security, and maintainability become central forms of relevance that image-like quality metrics cannot see.

7.1 Design Makes Relevance Falsifiable

Design is where the abstract theory of relevance becomes experimentally tractable. A designed object usually exists for users in a situation and must satisfy constraints. The chair must support weight, the interface must enable a task, the medical device must meet safety rules, the bridge must survive loads, the service must fit organizational processes. This makes it possible to distinguish hard gates from soft preferences in a way that art rarely permits.

Classical design-creativity research often combines novelty with usefulness or effectiveness. Shah, Smith, and Vargas-Hernandez proposed metrics for ideation that examine variety, novelty, quality, and quantity. The underlying intuition remains valuable in generative design: an unusual concept is not relevant if it fails the problem, while a perfectly functional concept may contribute little if it merely reproduces what already exists. Relevance appears in the relation between constraint satisfaction, user value, and meaningful departure from alternatives.

Design is closer to research than art in the respect that matters most for this book: there is usually a problem, a user, constraints, and criteria of functioning. Part of relevance can therefore be observed empirically. A chair can be uncomfortable, an interface can induce errors, a part can violate tolerances, and a service can be too expensive for its target user.

Falsifiability in design is rarely absolute. A usability test samples particular users; a stress test covers a defined load regime; a simulation relies on a model. The advantage is not certainty but a clearer link between claims and observations. Design evaluation becomes more rigorous when it records where a test applies and where it does not.

The creativity literature in design has tried to combine novelty with utility. Shah, Smith, and Vargas-Hernandez proposed metrics for ideation effectiveness, while the CPSS tradition discusses novelty, resolution, and elaboration/synthesis. These frameworks are remarkably compatible with the AI problem. A model can generate

extremely novel forms, but if they fail the brief, they are better understood as explorations than solutions.

Design benefits from explicit constraints because different claims can be tied to different tests. The table separates several common dimensions so that a strong result on one does not compensate invisibly for failure on another.

Dimension	Evaluating AI design without mixing criteria
Constraints	Mandatory conditions: safety, compatibility, dimensions, cost, regulation, and explicitly defined sustainability requirements.
Utility	Performance for real tasks and users, measured through tests and scenarios.
Novelty	Distance from existing solutions and avoidance of superficial variation.
Attractiveness	Preference and perceptual quality, ideally segmented across user groups.
Meaning	Authenticity, identity, symbolism, and relationship with a community or brand.
Portfolio	A set of diverse alternatives can be more valuable than the individually highest average score.

Table 7. Evaluating AI design without mixing criteria.

This separation also improves automated search. An AI system can explore Pareto trade-offs among safety, cost, usability, and performance rather than pretending that a single benchmark captures the whole design problem.

In design, relevance can be understood as the overlap among three spaces. The first is the constraint space: what cannot be violated. The second is the space of user needs and preferences. The third is the opportunity space: available alternatives, costs, materials, and technologies. An artifact is relevant if it occupies an interesting region of this intersection, not because it receives a high aesthetic score in isolation.

AI adds a new social dimension: the identity of the author can change evaluation. Recent work in fashion design suggests that products labeled as AI-designed may be judged less favorably, partly

through perceived authenticity and assumptions about expertise or intentionality. This is not necessarily a bias "error" that should simply be removed. For expressive products, the story of origin may be part of value. For strictly functional products, the same information may be less relevant.

A robust evaluator should therefore proceed in layers. Safety, legality, physical feasibility, interoperability, and mandatory dimensions are gates. A concept that fails them should not receive compensation from beauty. Functional performance is then measured through simulation, tests, task completion, or reliability. Human factors such as cognitive load, accessibility, ergonomics, trust, and learnability require user evidence. Cost and manufacturability introduce organizational constraints. Symbolic and aesthetic meaning can be assessed separately rather than smuggled into a general 'quality' score.

This layered structure makes design a good domain for testing AI evaluation architectures. Many claims can be falsified. A generated component either withstands a specified load in simulation or it does not. An interface either allows representative users to complete a task within an acceptable error range or it does not. A layout either meets accessibility rules or it does not. The evaluator can therefore learn which of its predictions correspond to real-world performance instead of relying only on human preference labels.

The larger lesson is that relevance becomes stronger when the artifact is connected to a testable theory of use. Even in domains where value is less objective, asking 'what would we observe if this artifact were relevant in the claimed way?' is a powerful discipline. Design gives us the clearest version of that question.

7.2 AI Design as a Research Programme

AI transforms design from a sequence of handcrafted proposals into a search process over enormous candidate spaces. A system can vary geometry, materials, interface structure, wording, workflow, visual hierarchy, and manufacturing parameters far faster than a human team can inspect them. Generation is therefore only the first stage.

The real design intelligence lies in deciding which simulations to run, which failures to investigate, which users to test, and when the evidence justifies convergence.

This makes AI design structurally similar to experimental science. The brief defines a problem and constraints. Generated variants act as hypotheses. Simulations and user studies produce evidence. Critics identify failure modes and hidden assumptions. The system updates its search based on what it learns. Importantly, the process can preserve provenance: why a variant was generated, which test eliminated it, which trade-off caused another to survive, and how later changes propagated through the design.

A design evaluator should therefore separate technical criteria from symbolic ones. It can simulate robustness, cost, and ergonomics; evaluate novelty relative to a corpus; and estimate preference. Authenticity, social meaning, and status may require models of communities rather than a single benchmark.

Generative engineering therefore needs evaluation budgets, not only generation budgets. If a system can create ten thousand candidates but can rigorously test only fifty, the intelligence of the workflow is concentrated in experiment selection. Which candidates are informative to test, which constraints are safety-critical, and when evidence is sufficient to stop searching become first-class design decisions.

Design is probably the first domain in which AI relevance infrastructure will become commercially indispensable. When a model can generate ten thousand concepts in an hour, value no longer lies mainly in generation, but in triage, simulation, diversity, and selecting a small portfolio of prototypes worth manufacturing.

Design evaluation can be formalized as a sequence of filters with different severities. Safety and physical constraints act as gates: if they are violated, a concept is rejected regardless of attractiveness. Utility and cost are usually optimization criteria. Meaning, novelty, and identity can be differentiating criteria. This structure is more robust than a weighted average because a high aesthetic score cannot compensate for a safety violation. The same pattern is useful

in project evaluation and AI artifacts: some dimensions are thresholds, others are trade-offs.

AI also enables adaptive relevance. Instead of designing for an "average user," a system can generate variants for subgroups and rare needs. This can increase local relevance dramatically, but it introduces questions of fragmentation, cost, and equity. A design personalized for every individual may be functionally superior yet destroy interoperability or exclude users for whom insufficient data exist. Individual relevance must therefore be held in tension with infrastructural relevance.

The most interesting benchmark for AI design would not be an image contest, but an end-to-end circuit: an unseen brief, generation of alternatives, simulation, selection, prototyping, user testing, and failure analysis. The evaluator would be judged by total cost to a valid solution and by the diversity of options preserved, not merely by how attractive the first proposal looks. This is exactly the kind of benchmark in which creativity evaluation begins to resemble automated scientific discovery.

Adaptive relevance introduces both opportunity and risk. AI can generate variants for different user populations rather than forcing everyone into an average persona. Accessibility can be designed in rather than patched later. Local regulations, languages, environments, and device constraints can be incorporated automatically. Yet extreme personalization can fragment products, complicate maintenance, and encode stereotypes if the system infers needs from demographic proxies rather than observed constraints.

A meaningful benchmark for AI design should therefore be end-to-end. The model receives an unseen brief with hidden or partially observable constraints. It generates candidates, chooses tests, interprets results, revises the design, and finally justifies the selected solution against baselines. Evaluation occurs not only on the final image or geometry but on robustness, constraint satisfaction, user performance, novelty relative to prior art, and the quality of the reasoning trace. Such a benchmark is much closer to real design work than aesthetic voting on isolated renders.

Design also reminds us that relevance can change after deployment. A product may be excellent in laboratory testing and fail because maintenance is difficult, procurement incentives differ, or users appropriate it in unexpected ways. The relevance dossier should therefore remain open after launch, accumulating failure reports, adaptation patterns, and evidence from real use. In this sense, good design evaluation is longitudinal science applied to artifacts.

7.3 Software, Infrastructure, and Maintenance Relevance

Software is often treated as easier to evaluate than art because tests can tell us whether code works. That is only partly true. Unit tests can verify specified behaviour, benchmarks can measure performance, and security tools can expose certain classes of weakness. But a software artifact becomes relevant inside an environment of users, dependencies, standards, institutions, adversaries, budgets, and future maintenance. A program that passes every benchmark may still be a bad engineering choice if it is impossible to operate, audit, integrate, or repair.

The first distinction is between functional correctness and system relevance. Functional correctness asks whether the artifact does what its specification requires. System relevance asks whether that specification corresponds to a real need and whether the solution remains credible under operational constraints. In research software this includes reproducibility, data provenance, environment capture, license compatibility, and the ability of other groups to understand and extend the code. In critical infrastructure it also includes incident response, supply-chain risk, graceful degradation, and governance over updates.

AI code generation makes this distinction urgent because it lowers the cost of producing implementations. A coding agent can generate several architectures, rewrite modules, add tests, and benchmark alternatives in minutes. The scarce competence shifts toward specification, threat modeling, evaluation design, architecture, and

evidence about maintainability. The best-looking implementation may be less relevant than a simpler one whose behaviour can be understood and bounded.

Security is a particularly strong example. Vulnerability counts are tempting metrics, but they are conditional on what the scanner knows how to find. An absence of reported vulnerabilities can mean robust code, poor testing, a small user base, or simply undiscovered defects. A relevance dossier for security-sensitive software should therefore keep separate evidence about design assumptions, attack surface, formal properties, test coverage, dependency provenance, red-team findings, operational history, and unresolved uncertainty. Compressing these into a single 'secure' score invites false confidence.

Interoperability has similar strategic importance. A technically superior component can have low practical relevance when it requires an ecosystem to be rebuilt around it. Conversely, a modest component that implements an important open standard may unlock value across many systems. This is why research infrastructure often produces indirect relevance: a dataset, protocol, library, or verification tool can enable hundreds of downstream projects without appearing as the headline result of any one of them.

Maintenance changes the time horizon. Software relevance decays when dependencies disappear, APIs change, security assumptions break, or the original team leaves. It can also increase when a standard gains adoption or a new application appears. Evaluation therefore needs longitudinal signals: issue resolution, contributor diversity, release discipline, backward compatibility, reproducible builds, dependency health, and the survival of knowledge beyond individual developers. Popularity metrics such as stars or downloads are informative about attention, not sufficient evidence of maintainability or criticality.

AI-generated infrastructure should be assessed through adversarial experiments rather than prose claims. Can another agent reproduce the environment from documentation? Can a red-team agent find a path that the design review missed? Can a migration

agent replace a dependency without changing semantics? Can the system explain why a security decision was made and which assumption would invalidate it? These tests turn abstract virtues such as transparency and resilience into observable challenges.

The same logic applies to scientific software and automated research pipelines. If an AI system produces an analysis, the relevant artifact is not only the final graph or conclusion. It includes code, data transformations, model versions, prompts or policies, intermediate decisions, and provenance. Relevance to science depends on whether another investigator can inspect the path from evidence to claim. A fluent narrative that cannot be audited is less relevant than a slightly rougher result with a reproducible chain of reasoning.

This is where design and research converge most strongly. Technical artifacts make claims about the world: that a bridge will carry a load, that a protocol will preserve a property, that an interface will reduce errors, that an algorithm will generalize, that a deployment will remain within a budget. AI can generate candidates, but relevance emerges only when those claims are tested against the constraints that matter.

Benchmarking deserves particular caution because AI systems can optimize against benchmarks faster than institutions can redesign them. A coding model that knows the public test suite may learn to satisfy the measured cases without acquiring the broader competence the benchmark was intended to represent. The engineering analogue of Goodhart's law is familiar from security competitions, machine-learning leaderboards, and performance tuning: once the target is visible, effort migrates toward what is scored. Relevant evaluation therefore needs hidden tests, distribution shifts, adversarial cases, and periodic replacement of saturated benchmarks.

For infrastructure, the most important property may be reversibility. Organizations should ask how difficult it is to migrate away from a component, reconstruct decisions, export data, replace a model, or recover after a maintainer disappears. These are rarely

glamorous metrics, yet they determine whether short-term technical success becomes long-term institutional dependence. AI can help estimate such risks by simulating migration and failure scenarios, but the criteria must be defined before the convenience of the current system becomes the default standard.

Technical relevance is also collective. A library can be locally inferior to a custom implementation and still be globally valuable because many teams share its maintenance and verification costs. Open standards, common interfaces, and reusable infrastructure create option value for ecosystems. This resembles portfolio relevance in funding: the best component in isolation is not always the component that makes the surrounding system strongest.

Once technical creation becomes cheap, organizations will need to value architectures that remain understandable under change. The next chapter shows why art pushes in the opposite direction: some valuable artifacts are important precisely because they resist stable specifications. A general theory of relevance must accommodate both cases without pretending they are identical.

Design and software make clear that relevance can be operational without being simplistic. Tests, simulations, benchmarks, and audits are valuable because they connect claims to evidence, but only if their scope is explicit and the system remains open to failures outside the benchmark.

Art now provides the complementary limit case: a domain where part of the value may lie in changing the benchmark itself.

8. Visual Art: Relevance Beyond Liking

Visual art is where a universal relevance metric most visibly breaks. A system can estimate visual quality, prompt adherence, stylistic similarity, or average preference, but important art has repeatedly violated the criteria by which its contemporaries expected art to succeed. A work can be ugly by prevailing standards and historically important; technically simple and conceptually generative; commercially ignored and culturally durable.

This does not mean art is beyond evidence. It means the evidence is plural. Institutions can observe interpretation, influence, community significance, formal novelty, provenance, preservation, and historical reuse. What they cannot honestly do is pretend these signals are interchangeable or that one model has discovered the final latent variable called artistic value.

AI-generated art adds a second complication: people react not only to images but to beliefs about who or what made them. The chapter uses this as a case study in why provenance can be part of relevance and why evaluation systems must distinguish contextual meaning from perceptual preference.

8.1 Why Liking Is Too Small a Measure

Visual art draws a hard boundary around what preference models can legitimately claim. Modern image models can estimate aesthetic appeal, technical quality, prompt adherence, or similarity to a corpus. Those signals are useful for production systems. They are not equivalent to artistic relevance because a work may matter precisely by frustrating ordinary expectations of beauty, finish, harmony, or immediate intelligibility.

The AHRC Cultural Value Project is useful here because it treats cultural value as a plural phenomenon involving individual experience, social connection, identity, civic effects, and other dimensions that cannot be collapsed into market value. This perspective helps separate the question 'Do people like this image?' from 'What does this work do within a cultural context?' The second question requires interpretation, history, institutions, and sometimes conflict.

Visual art is the case that most strongly limits the temptation to transform relevance into preference prediction. Models such as NIMA can estimate distributions of aesthetic scores and correlate well with human judgments of image quality. HPS and HPS v2 learn human preferences for text-to-image systems and are excellent reward models for producing images users prefer. But a preference reward model is not an art critic, and still less a cultural historian.

Aesthetic models such as neural image assessment systems are useful when the target is explicitly perceptual quality or preference. Problems arise when these models are silently promoted into theories of artistic value. Their success on rating prediction should be understood as task performance within a defined label regime, not evidence that art has been reduced to computable taste.

The reason is simple. A relevant work can be unpleasant, difficult to understand, technically imperfect, or initially rejected. Sometimes relevance is produced by tension with convention. If we optimize exclusively for aggregate preference, we learn the center of the taste

distribution and penalize precisely the deviations that might change that distribution in the future.

Automated visual assessment is valuable when its target is clearly stated. The table contrasts measurable perceptual or corpus-based signals with broader artistic claims that require context and interpretation.

Dimension	What visual models measure — and what they do not
Aesthetic score	Can approximate human preferences over visual quality and aesthetics within the training distribution.
Prompt alignment	Measures how well an image follows an instruction; does not establish cultural value.
Human preference	Useful for ranking and reward; may favor median taste and existing conventions.
Novelty	Can be approximated through distance from a corpus, but distance is not equivalent to artistic contribution.
Cultural relevance	Requires tradition, context, communities, interpretation, and time; no universal defensible benchmark exists.
Authorship context	In expressive arts, authorship can be part of the evaluated object rather than an external “bias” to be automatically removed.

Table 8. What visual models measure - and what they do not.

The aim is not to place art beyond analysis. It is to stop technical success on one prediction task from acquiring authority over a larger cultural question that the model was never trained to answer.

Research on AI-generated art also shows that evaluation depends on how origin is labeled. People may assign less value to an image when they believe it was generated by AI, and the effect varies with product type and perceptions of intention, authenticity, or threat. From an engineering perspective, it is tempting to call all of this “bias.” From the perspective of art, some of it is context. If a work is valued partly as an expression of intention or biography, changing the author changes the evaluated object.

The AHRC Cultural Value Project offers a broader lesson. Cultural value is not reduced to economic impact; it includes individual experience, capacity for reflection, social relationships, identity, and community effects. Measurement must accept methodological pluralism. This position fits surprisingly well with responsible research assessment: in both cases, the valuable object is too rich for a single indicator.

Empirical work on AI-generated art also reveals a source-attribution effect. Studies in 2024 and 2025 report that people often evaluate artworks less positively when they believe the works were produced by AI, even when perceptual engagement or image quality is comparable. Cunningham and colleagues, for example, found more negative subjective judgments when participants were told religious artworks were AI-generated, while eye-movement measures did not show corresponding differences in visual exploration. This does not tell us which judgment is correct. It tells us that perceived authorship is part of the evaluative context.

That fact complicates automated assessment. If a model is trained on human ratings without recording whether raters knew the source, it may learn a mixture of visual preference and social beliefs about authorship. Similar confounding appears with prestige labels, gallery context, price, artist biography, or institutional endorsement. An artistic relevance system should therefore preserve contextual variables rather than treating ratings as pure observations of the image.

The central distinction is between aesthetic preference and cultural argument. Preference can be measured comparatively and is valuable for recommendations or creative tools. Cultural relevance requires evidence about relation to traditions, symbolic meaning, interpretive effects, communities, and historical position. The latter will remain partly contestable, which is not a reason to abandon evaluation; it is a reason to represent disagreement honestly.

8.2 Art Can Change the Evaluator

Art is exceptional because it can alter the criteria by which later art is judged. A scientific claim becomes stronger when evidence confirms it under stable epistemic rules. An artwork may become important by violating the aesthetic rules that initially make it look wrong. This reflexive capacity means that an evaluator trained only to reproduce current expert consensus can become structurally conservative even if its average judgments correlate well with experts.

Canon formation makes the same point institutionally. Museums, critics, collectors, schools, publishers, markets, archives, and political movements influence which works survive long enough to be reconsidered. Historical relevance is therefore partly observed and partly produced. A model trained on surviving canons will learn a selection process that contains both artistic achievement and historical exclusion. Without explicit safeguards, it can present that mixture as timeless taste.

For AI art, relevance could be assessed through a dossier that includes positioning within tradition, formal novelty, interpretive density, responses from diverse communities, relation to the historical moment, and the capacity to generate new practices. None of these dimensions must necessarily be numerical. Some are better expressed through comparisons, counterexamples, and critical argument.

The reflexive character of art also appears in science, though more weakly: revolutionary theories can change what later scientists consider a good explanation. Art makes the phenomenon impossible to ignore. A relevance infrastructure that cannot revise its own criteria is therefore unsuitable for domains in which important artifacts can create new evaluative vocabularies.

Art therefore forces us to accept an idea that is central to this book: good evaluation does not always maximize certainty. Sometimes its role is to identify what cannot yet be reduced but deserves to remain in view. A mature relevance system needs a

category such as "promising but uncalibrated," together with mechanisms for re-evaluation over time.

It is useful to distinguish four types of artistic relevance without pretending to exhaust aesthetics. Internal relevance concerns coherence and tension within the work. Historical relevance concerns its relationship with tradition and the moment in which it appears. Social relevance concerns the communities, conflicts, and identities it touches. Generative relevance concerns what the work makes possible: new styles, new interpretations, new practices. The last is especially important in the AI era because a work may be valuable not through immediate preference, but because it shifts the creative space for subsequent works.

A preference model usually learns from comparisons of the form A versus B. Such data are extremely useful for aligning a product with user taste, but they turn the question "which of these two images do you prefer now?" into a substitute for "which work deserves to exist?" The difference is enormous. The second question includes time, influence, risk, and minority value. In a mature system, preference models should be one sensor among others, not the supreme court.

Art also provides an argument for adversarial evaluation. One critic can deliberately try to show that the work is derivative; another can search for the strongest possible interpretation; a historian can test its relation to tradition; a community evaluator can examine local resonance. Instead of asking one LLM to play all these roles simultaneously, we can preserve the verdict as a structure of arguments and disagreements. This architecture resembles real critical deliberation more closely and reduces the risk that the model's average taste becomes a new invisible cultural institution.

One response is adversarial plurality. Instead of one model assigning a final artistic score, several critics can take different roles. One searches for derivative structure and close precedents. Another argues for formal innovation. Another examines cultural and political context. Another tests how the interpretation changes across communities. A final synthesizer records agreements and

tensions without forcing a single verdict where the evidence remains value-dependent.

Another response is temporal humility. The system can distinguish current relevance from option value. Current relevance asks whether the work has a defensible place in today's conversation. Option value asks whether there are reasons not to eliminate it even if current consensus is weak. High formal novelty, unusual combinations of traditions, or strong disagreement among informed critics may justify preservation for later reevaluation. Cultural institutions already do this implicitly when archives keep works that are not fashionable.

This is the deepest lesson art contributes to a general science of relevance. Evaluation should sometimes protect objects from evaluation. A system optimized to remove everything that looks weak under present criteria may become efficient at destroying the raw material from which future criteria could emerge. Relevance therefore requires an exploration reserve: space for artifacts that are not yet legible to the dominant model of value.

8.3 Cultural Value, Authorship, and Preservation

The cultural-value literature provides a useful antidote to the idea that artistic relevance can be measured by market price or audience size. The AHRC Cultural Value Project, led by Geoffrey Crossick and Patrycja Kaszynska, synthesized evidence across individual experience, social connection, identity, civic life, health, education, and economic effects. One of its important lessons is methodological: cultural value is not a hidden single quantity waiting for the correct instrument. Different forms of value appear through different mechanisms and require different kinds of evidence.

This plurality is especially important for AI-generated images because human judgments depend on what observers believe they are looking at. Grassini and Koivisto found a negative bias toward artworks perceived as AI-generated, with attitudes toward AI helping explain individual differences. Later studies reported related

authorship-attribution effects. Such findings should not be dismissed as contamination that a perfect evaluator would remove. In art, provenance can legitimately be part of meaning. The mistake would be to mix provenance effects with visual properties and then report the result as a pure aesthetic score.

Consider two visually identical images, one produced as the final work of an artist documenting a personal history and the other generated accidentally during a model sweep. Their perceptual properties can be identical while their cultural functions differ. A museum may reasonably care about the chain of intention, testimony, authorship, and institutional context. A wallpaper recommender may reasonably ignore most of that context. Both judgments can be coherent because they answer different questions. Relevance is relational again.

The difficulty increases when AI is used as part of human practice. Authorship may be distributed across prompt design, dataset curation, model choice, iterative selection, editing, physical fabrication, and institutional framing. Trying to assign a single percentage of authorship can become less informative than documenting the actual process. For evaluation, process provenance allows critics to ask whether the work's meaning depends on a specific human intervention or whether similar outputs emerge routinely from the generator.

Preservation creates another asymmetry. It is expensive to place a work in front of millions of people but relatively cheap to preserve a digital record for future discovery. Cultural institutions can therefore separate current promotion from long-term option value. An artwork that is difficult to interpret today may deserve archival survival because it occupies an unusual formal or cultural position. The purpose is not to declare it secretly great; it is to avoid making irreversible decisions with weak evidence.

This suggests that museums, archives, and cultural platforms should measure coverage of possibility space as well as present consensus. If acquisition or recommendation algorithms consistently favour works close to established canons, the system may become

efficient at reproducing yesterday's legibility. An explicit exploration budget can reserve attention and preservation capacity for unfamiliar clusters, minority traditions, new media, and works that create unusually strong disagreement among informed evaluators.

Disagreement itself is not proof of relevance. Provocation can be manufactured cheaply, especially when generative systems can search for outrage. The important distinction is between arbitrary disagreement and interpretively productive disagreement. A work has stronger evidence of the latter when critics can articulate different coherent readings, when the work reveals assumptions in existing categories, or when later works and discussions use it as a reference point rather than merely as an object of controversy.

An AI evaluator can assist this process by retrieving precedents, tracing iconographic or stylistic similarities, checking factual claims about context, and simulating several critical perspectives. It should be much more cautious about final aesthetic authority. Models trained on cultural corpora inherit the survival biases of those corpora: what was collected, digitized, described, translated, and canonized. A system can reproduce those patterns with impressive confidence while silently narrowing the future.

For this reason, the most valuable output of automated art assessment may be a structured disagreement map. One critic argues formal novelty, another derivative structure; one examines historical context, another community meaning; one asks what disappears if provenance changes; another looks for evidence that the work has generated new practice. A human curator can then inspect the tensions rather than receiving a synthetic number that erases them.

Generative abundance also changes the economics of cultural preservation. Physical museums, galleries, and archives must make hard acquisition decisions because storage, conservation, and exhibition are expensive. Digital artifacts can be preserved much more cheaply, but description, metadata, rights management, and future discoverability still require work. A relevance system should therefore distinguish a low-cost preservation threshold from a much higher promotion threshold. This allows institutions to protect

option value without pretending every stored artifact deserves equal attention today.

The archival problem becomes especially important when generative systems produce families rather than single works. Millions of near-variants are not meaningful cultural diversity. Preservation may need to operate on representative clusters, production traces, and unusually distant examples rather than raw volume. In that setting, relevance is partly a compression problem: preserve enough structure that future researchers can reconstruct the creative space without storing or foregrounding every redundant sample.

Art contributes the most demanding test for a science of relevance because it demonstrates that evaluation criteria can themselves be historical objects. The next chapter takes that lesson back to AI creation more generally: when generators can explore huge spaces, selection systems must learn without becoming prisons made of yesterday's preferences.

Art forces a final conceptual concession: some relevance cannot be fully known at selection time because the artifact may help create the interpretive context in which it later matters. Preservation and exploration are therefore not optional cultural luxuries; they are defenses against irreversible false negatives.

The next chapter generalizes this idea to AI creation as a whole and asks what happens when generation becomes an explicit search process.

9. When Creation Becomes Research-Like

The previous chapters appear to describe very different worlds. Science seeks warranted knowledge; literature organizes language and experience; film combines cultural and industrial value; design tests artifacts against constraints; art can transform the criteria used to judge it. Yet generative AI reveals a structural similarity beneath these differences. When candidate production becomes cheap, intellectual work increasingly takes the form of search through a space of possibilities.

Search alone is not intelligence. The decisive capabilities become asking which region to explore, choosing discriminating tests, interpreting evidence, preserving useful failures, and deciding when to stop. This is why advanced creative systems begin to resemble research workflows even when their final products are not scientific claims.

The chapter develops this convergence without collapsing art into science. What becomes research-like is the process of disciplined exploration, not the final criterion of value.

9.1 Generation Becomes Search

As generative systems become capable of producing text, code, designs, simulations, images, and plans, a structural convergence appears across intellectual work. Research has always been a search process: formulate a question, generate hypotheses, gather evidence, criticize explanations, compare alternatives, and update beliefs. When generation is expensive, other creative fields can hide this structure inside individual craft. When generation becomes cheap, the search structure moves to the foreground.

Consider software engineering. An AI agent can propose several architectures, implement prototypes, run tests, benchmark latency, inspect vulnerabilities, and revise the design. The useful output is not the first code sample but the sequence of hypothesis and falsification. Consider policy analysis: a system can generate intervention options, retrieve comparable cases, simulate stakeholder responses, identify legal constraints, and stress-test causal assumptions. Consider a book: the system can explore outlines, test arguments against literature, solicit critics with different perspectives, and refine the manuscript. These are research-like workflows even when the final artifact is not a scientific paper.

As AI advances, the boundary between research and other forms of intellectual work will blur for a structural reason. Research is, at its core, a loop of hypothesis generation, evidence search, artifact construction, critique, experiment, and selection. The same loop begins to appear in software, strategy, design, policy, education, and content creation when generation becomes cheap and evaluation can be partially automated.

This is intentionally not a complicated agent architecture. The diagram captures the minimum loop required for creation to become research-like: alternatives are generated, subjected to tests, criticized, selected, and used to update the next search.



Abundant generation moves intelligence toward evaluation and iteration.

Figure 4. Abundant generation moves intelligence toward testing and selection.

If the learn step does not change generation or evaluation, the process is only high-volume sampling. The loop becomes intelligent when evidence can redirect it.

Search is not a metaphor for merely producing many variants. A genuine search process has a state, an objective or family of objectives, evidence from previous trials, and a policy for deciding what to explore next. When AI systems simply sample thousands of candidates and rank them with the same frozen model, they can amplify the evaluator's blind spots at scale. Research-like creation requires feedback capable of changing the search itself.

An agent designing a new interface can generate dozens of variants, simulate use, analyze benchmarks, consult accessibility rules, identify failures, and revise the design. An agent writing a book can search the literature, construct a taxonomy, generate narrative hypotheses, ask specialized critics to find banalities, and produce alternative versions. A software system can generate implementations, run tests, analyze regressions, and compare architectures. From the perspective of cognitive workflow, all of these become miniature research programmes.

The table translates the chapter's argument into process terms. Each stage becomes a site at which evidence can improve the next decision rather than merely a step toward producing more output.

Stage	From creation to a research-like workflow
Problem framing	Define the situation, values, and criteria before generation.
Search / generation	Produce a broad space of hypotheses, texts, designs, or solutions.
Evidence / simulation	Confront artifacts with data, rules, users, models, and counterexamples.
Critique	Separate evaluator roles rather than relying on one omniscient judge.
Selection	Choose under multiple criteria while preserving diversity, not merely maximizing average score.
Learning	Compare relevance predictions with later outcomes and recalibrate evaluators.

Table 9. From abundant creation to a research-like workflow.

The crucial transition is from ranking static candidates to learning from tests and criticism. Without that feedback, scale merely multiplies the biases of the evaluator used to select among generated artifacts.

This researchification of creation has an economic consequence. If every organization can produce thousands of candidate artifacts, value shifts toward three functions: defining good problems, constructing enough context for evaluation, and maintaining a selection process that does not close itself too early. The first function is axiological, the second epistemic, the third institutional.

Art remains partly exceptional because it can change the criteria by which it is evaluated. Yet even there the production process can become research-like: exploration of an aesthetic space, testing reactions, comparison with a corpus, searching for surprising combinations, and portfolio selection. The difference is that the final verdict cannot be reduced to falsification or utility.

The key word is 'like.' The convergence concerns process, not the final criterion of truth. A novel can be produced through systematic experimentation without becoming science. A design can be investigated experimentally while still requiring aesthetic and symbolic judgment. A film can be optimized through audience

studies without cultural value becoming reducible to those studies. The research-like workflow gives creators better ways to search and learn; it does not erase domain-specific values.

This shift changes where intelligence is required. If the generator can produce thousands of alternatives, marginal improvement in generation may matter less than better problem framing, evidence acquisition, critique, and stopping rules. The most valuable agent may be the one that knows which experiment to run next, which disagreement deserves escalation to a human, or which attractive direction should be abandoned because the evidence is weak.

Relevance becomes the coordinating variable across the workflow. It decides which candidates enter testing, which evidence matters for which claim, which failures are fatal, which uncertainties justify exploration, and which survivors belong in the final portfolio. Generation produces possibility; relevance turns possibility into a programme of work.

9.2 Authorship Moves Up a Level

The human role changes accordingly. Traditional authorship often equates responsibility with direct production: the author wrote the sentences, the designer drew the form, the programmer wrote the code. In an AI-intensive process, a person may contribute less local execution while exercising more control over the search space. They choose the question, define constraints, select evidence, appoint critics, decide which values are non-negotiable, and accept responsibility for the final selection.

This resembles the role of a principal investigator or laboratory director. A PI does not personally perform every experiment, yet remains responsible for the research programme: why the question matters, whether methods are appropriate, how contradictory evidence is handled, and when claims are ready. The future author of an AI-generated intellectual artifact may similarly be responsible for the epistemic and axiological programme behind generation rather than for every token or pixel.

The greatest mistake would be to use the same relevance function for all these domains simply because the same family of models generates their artifacts. Uniformity of production technology does not imply uniformity of value. On the contrary, the more kinds of content the same AI can produce, the more explicit the evaluation system must be about the ontology of each domain.

This does not make human authorship disappear. It relocates part of authorship from direct fabrication to the design of questions, constraints, evaluators, experiments, and stopping rules. A person who defines what evidence can overturn a preferred direction may exercise more substantive agency than one who manually edits every sentence after an opaque generator has already determined the conceptual space.

The future will therefore not be dominated only by larger generative models. It will also be dominated by evaluation infrastructures: critics, simulators, knowledge graphs, stakeholder models, provenance systems, longitudinal benchmarks, and portfolio mechanisms. In a world of nearly unlimited creation, the most valuable agent may be the one that knows what not to generate next.

This convergence between creation and research does not mean that all domains become science. It means that their production processes acquire an experimental form. When variants are cheap, it becomes rational to formulate hypotheses about audiences, generate alternatives, measure reactions, search for counterexamples, and revise. Even a novel can have a research pipeline around it without the value of the novel being reducible to data. The important distinction is between an epistemic workflow and a final criterion of value.

The role of the author also changes. In a massive generative process, the author may become less the executor of every sentence and more the constructor of the search space: defining the problem, selecting sources, creating constraints, deciding which critics are legitimate, and deciding when the process should stop. This resembles the role of a principal investigator or laboratory director. Authorship does not disappear; it moves from local production of

every token toward responsibility for the programme of exploration and selection.

In this scenario, institutions that own the best generators do not necessarily hold the decisive advantage. Generators tend to commoditize. Advantage may belong to those who possess hard-to-copy evaluation contexts: longitudinal data, expert communities, real feedback from use, simulation infrastructure, and good models of consequences. The economic moat can move from the model that produces to the system that knows what is worth trying and can demonstrate why.

That higher-level authorship also creates new provenance requirements. A meaningful record should show which sources entered the process, which parts were generated, which tests were executed, what human interventions changed direction, and which evaluators influenced selection. Provenance is not merely a copyright feature. It is part of relevance because it allows later readers to understand why the artifact survived the search process and which assumptions shaped it.

Economically, this suggests that generation itself may commoditize faster than evaluation context. Many organizations may have access to similar frontier models, but far fewer will possess expert communities, proprietary feedback loops, longitudinal outcome data, trustworthy simulators, high-quality benchmarks, or institutional credibility. The durable advantage may move from 'we can generate' to 'we can demonstrate why this is worth pursuing.' In that world, evaluation infrastructure becomes a competitive moat.

The change is not a demotion of human creativity. It is a redistribution of effort. Humans can spend less time producing low-level variants and more time on framing, judgment, ethical responsibility, and the creation of new criteria. That is also why transparent relevance systems matter: if humans are to remain responsible at a higher level, they need tools that expose rather than conceal the reasoning that shaped the final artifact.

9.3 Evaluation Changes the Thing It Measures

There is a second reason a general theory of relevance must go beyond static scoring: evaluation is performative. A recommendation system does not merely discover what a person likes; repeated recommendations shape what the person encounters and may therefore shape taste. A research programme does not merely measure which fields are important; its funding helps create laboratories, datasets, careers, conferences, and future citation networks. A literary prize does not merely reveal a canon; it contributes to building one.

This feedback loop becomes stronger when AI-mediated evaluation operates at scale. Suppose a platform learns that users engage most with a particular narrative structure and begins selecting or generating more of it. The training data for future models then contain more examples of that structure, and the system becomes increasingly confident that the structure is what audiences want. Local relevance can become cultural homogenization. In science, a novelty detector trained on historically funded proposals can similarly reinforce yesterday's definition of plausibility.

Relevance also has a reflexive property in culture. Evaluation does not merely measure value; it helps produce it. A recommendation algorithm can turn an obscure book into a phenomenon. A jury can consecrate a film. A grant can create a field by funding its infrastructure and people. When AI becomes an evaluator at scale, it will not be merely a thermometer of relevance, but one of the mechanisms by which relevance is distributed. Transparency of criteria therefore becomes a problem of cultural governance.

The danger becomes more acute when the generator can see or infer the evaluator. A model asked to maximize a fixed relevance score will discover linguistic and structural shortcuts that exploit it. The relevant comparison is adversarial security rather than ordinary prediction: the evaluator must remain meaningful under optimization pressure from systems specifically designed to satisfy it.

Multi-agent systems do not solve this merely by multiplying opinions. If all critics share the same training distribution, reward model, retrieval source, and implicit cultural assumptions, an ensemble can create the appearance of pluralism while remaining epistemically monocultural. Diversity of evaluators should therefore be designed around different evidence sources, methods, failure models, and objectives, not just different role prompts.

Experiment selection is the other underappreciated capability. When a generator can produce thousands of candidates, the best next test is often not the test of the current favorite. It may be the test that most sharply separates competing explanations or reveals whether an entire cluster of candidates shares the same hidden failure. This is the logic of active learning and scientific experimentation applied to creative search: spend scarce evaluation effort where it changes the decision most.

A general taxonomy can separate at least five meanings of relevance. Epistemic relevance concerns contribution to what can be known. Instrumental relevance concerns the ability to help achieve a goal. Experiential relevance concerns value as lived by an audience. Cultural relevance concerns changes to the language, memory, and norms of a community. Systemic relevance concerns the effect on the portfolio or ecosystem into which an artifact enters. The same artifact may occupy completely different positions on these axes. A documentary may be aesthetically modest but civically important; a mathematical model may be immediately useless but epistemically decisive; a design may be culturally elegant and instrumentally unusable.

Software shows this plurality clearly. For a software library, relevance may mean solving a frequent problem, reducing a security risk, becoming infrastructure for thousands of projects, or introducing an abstraction that changes programming practice. Stars and download counts are attention. More important signals may include transitive dependencies, supply-chain criticality, substitutability, and migration cost. For an AI agent that generates software, relevant evaluation looks more like research and

engineering assessment than text evaluation: claims can be executed, tests can falsify promises, and provenance matters.

Policy briefs, contracts, educational materials, and business plans complete the picture. Here truth and contextual fit matter greatly, but effect depends on the actor. A policy brief is relevant if it reaches the right decision-maker within a decision window and offers an actionable alternative. Educational material is relevant if it changes understanding for a learner at a particular level, not merely if it is correct. A business plan is relevant if it connects market assumptions to tests and signals that could falsify the plan. In every case, AI pushes us from evaluating a static product toward evaluating a theory of use.

The evaluator therefore has a dual responsibility. It must estimate relevance for the present decision, and it must account for the distributional effects of repeatedly applying the same criterion. This is where systemic relevance enters. Selecting the individually highest-scoring artifact every time may reduce diversity, resilience, option value, and the chance of discovering new categories. Portfolios need some degree of redundancy removal and some deliberate preservation of difference.

False negatives deserve special attention because they often remain invisible. A mediocre funded project can later be observed and criticized. A rejected proposal that would have opened a new field leaves no direct counterfactual trace. A book never shown to readers cannot prove that it would have found an audience. A design eliminated by a preference model cannot demonstrate that users would have learned to value it. Systems optimized only on observed successes systematically under-measure what selection prevented from existing.

For this reason, relevance evaluation is also governance. Criteria allocate attention, capital, legitimacy, and opportunity. The technical architecture must therefore preserve contestability: the ability to inspect why a decision was made, identify which criterion dominated, compare alternative value systems, and reserve some resources for exploration. The evaluator should not only answer

'what looks relevant now?' It should also ask 'what future distribution of possibilities will this rule create if we apply it repeatedly?'

Research-like creation does not mean that every poem needs a hypothesis test or every design needs a journal article. It means that high-level intellectual agency increasingly resides in the architecture of exploration: questions, constraints, critics, experiments, memory, and stopping rules.

Once authorship and intelligence move to that level, relevance evaluation becomes part of the creative system itself. The final chapter therefore treats evaluation as infrastructure.

10. Building Relevance Infrastructure

If relevance is relational, multidimensional, temporal, and vulnerable to gaming, what should an AI evaluation system actually produce? The wrong answer is a number presented with enough decimal places to look scientific. A scalar may be necessary at the final decision boundary, but it should be the visible tip of a richer structure rather than the substitute for one.

The final chapter proposes a relevance dossier: a structured record of the decision context, claims about why an artifact matters, evidence for and against those claims, comparable alternatives, uncertainty, disagreements among specialized critics, portfolio effects, and conditions under which the judgment should be revised. The system is designed to support human and institutional decisions, not to claim metaphysical authority over value.

This architecture also creates a research programme. Its components can be falsified separately: retrieval quality, novelty estimates, causal-chain checking, calibration, bias detection, critic diversity, longitudinal prediction, and resistance to gaming. A science of relevance becomes possible only when the system can learn from its own mistakes.

10.1 The Relevance Dossier

An auditable relevance system should begin before the first model call. Its first object is a context specification: what decision is being made, who is affected, which values and constraints are legitimate, what time horizon matters, which errors are costly, and which criteria are hard gates rather than trade-offs. Without this context, the model is forced to infer the institution from surface language, which is exactly how hidden value assumptions enter an apparently neutral score.

The second object is a claim graph. Instead of asking an LLM, 'How relevant is this proposal?', the system decomposes the judgment into explicit claims such as 'the problem is material for target group X,' 'the proposed method differs from the closest prior work in Y,' 'the consortium can plausibly execute task Z,' or 'adoption requires assumption A.' Each claim can then be linked to evidence, counterevidence, uncertainty, and an evaluator responsible for that dimension.

The dossier also creates a clean boundary between evidence and policy. Evidence can show that one proposal is more novel, another has stronger user validation, and a third has lower execution risk. It cannot decide how much an institution ought to value novelty relative to risk unless that preference is supplied. Encoding weights or thresholds explicitly is therefore a feature, not a limitation: it exposes the political and strategic layer that opaque scoring systems often disguise as technical intelligence.

The architecture keeps context and evidence upstream of judgment and forces specialized critics to leave inspectable traces before a dossier reaches a decision process.



The decision remains traceable to evidence and disagreement.

Figure 5. A simple architecture for auditable relevance assessment.

A decision-maker can still request a rank or score. The important property is reversibility: the rank can be traced back to assumptions and reconsidered when evidence or institutional priorities change.

A dossier is deliberately richer than a score because it separates evidence from decision policy. The same evidence can support different decisions in a frontier-research fund, a safety regulator, a publisher, or a recommender system. By preserving the evidence graph, assumptions, and uncertainty, the system allows policy to change without pretending that history itself changed.

If relevance is becoming infrastructure, the next question is technical: how should such a system be built? The first principle is that it should not begin with a score. It should begin with a context object: who is evaluating, for which decision, which objectives and values are declared, which constraints are mandatory, which time horizon matters, and which type of error is more costly.

The system proposed here is intentionally modular. The table shows the output expected from each component, which makes it possible to test failures separately rather than treating the whole evaluator as an inscrutable black box.

Component	The output of a good relevance system
Context	Decision, actors, values, constraints, and time horizon for which the assessment is valid.

Component	The output of a good relevance system
Claims	Separate statements about fit, novelty, quality, utility, probable impact, and risks.
Evidence	Sources, comparisons, tests, simulations, and traceability for each important claim.
Uncertainty	Intervals, evaluator disagreement, and assumptions on which the verdict depends.
Alternatives	Which other options were compared and under what conditions they become preferable.
Revision trigger	Future events or data that should trigger re-evaluation.

Table 10. Components of an auditable relevance system.

Modularity also supports institutional variation. A frontier-research fund, a publisher, and a design-review board can use different critic modules and decision policies while sharing the same discipline of evidence, provenance, uncertainty, and revision.

The second principle is decomposition. A novelty evaluator should not be the same module that decides credibility. A call-fit module should not determine societal value. A bias critic should be able to challenge a preference evaluator. Separation enables audit, comparison of models, and replacement of components without rebuilding the entire system.

The third principle is evidence linking. Every important evaluative claim should be able to point to the source, comparison, or test that supports it. In proposal review, "high novelty" should be linked to a search of relevant literature and examples of close work. In design, "high utility" should be linked to tests or simulations. In art, "an innovative relationship with tradition X" should be argued through comparisons rather than the internal probability of a model.

The fourth principle is plurality. Instead of a single LLM evaluator, the system should combine specialized perspectives and preserve disagreement. LLM-as-a-judge is useful, but recent literature documents position bias and other sensitivities. If the order of two

answers can change the verdict, that fact should not be hidden behind a decimal. Repetition stability, position consistency, and preference fairness are examples of tests that should be applied to the evaluator itself.

Evidence should be typed. A peer-reviewed paper, benchmark result, simulation, policy document, market report, user study, expert opinion, and model-generated inference are not interchangeable. The dossier records what kind of evidence supports each claim and whether the evidence is direct, indirect, historical, or speculative. This simple discipline reduces one of the most common failures of LLM evaluation: fluent synthesis that erases the epistemic differences among sources.

Alternatives belong inside the dossier rather than in an appendix. Relevance is comparative, so the system should state which baseline, competing proposal, prior artwork, existing product, or no-action scenario was used. An intervention that looks impressive in isolation may become less relevant when a cheaper alternative is considered. Conversely, an incremental result may be highly relevant if it uniquely removes a bottleneck that no other option addresses.

The final component is a revision trigger. Every assessment should state which future observation would cause reevaluation: a new competing technology, failure of a key assumption, evidence of adoption, a regulatory change, replication failure, or the emergence of an unexpected audience. A dossier that cannot be revised is only a decorated score. A dossier that records the conditions of its own invalidation can participate in a learning system.

10.2 A System of Critics, Not an Oracle

The evaluator should be a system of critics rather than a single omniscient model. Specialization makes both reasoning and benchmarking easier. A novelty critic searches for close prior work and distinguishes substantive difference from rhetorical novelty. A feasibility critic inspects dependencies, resources, and failure modes. A relevance-to-call critic maps proposal claims to official requirements. A user-value critic examines evidence about

beneficiaries. A cultural critic interprets relation to traditions and communities. An adversarial critic tries to break the arguments produced by all the others.

This architecture is not automatically trustworthy merely because it contains several agents. Multiple models can share the same bias, cite the same mistaken source, or converge because they are prompted similarly. Diversity should therefore be engineered through different models, retrieval sources, role instructions, methods, and where appropriate human evaluators. The system should measure correlation among errors, not simply count votes.

Specialization is also a defense against plausible nonsense. One critic can test novelty against literature, another can inspect causal logic, another can look for methodological failure, another can test policy or call alignment, and another can search for counterexamples. Their disagreements are information. A final synthesizer should not erase them before the decision-maker has a chance to see where the case is fragile.

The fifth principle is portfolio reasoning. The relevance of an artifact depends partly on what has already been selected. If ten nearly identical projects have been funded, the eleventh may have lower marginal relevance even if its individual score is high. If only similar films are recommended, local relevance may increase while systemic value declines. Pareto fronts, diversity constraints, and exploration budgets are more appropriate than a raw top-N list.

The sixth principle is longitudinal audit. Any system that predicts relevance should be judged by what happened after the decision. Were results adopted? Did the impact appear? Did an apparently marginal idea become a sleeping beauty? Did selection produce bias against a group, discipline, or new form? Without this loop, "AI relevance" risks becoming automated rhetoric.

The final output should be a relevance dossier: a structured representation containing dimensions, evidence, uncertainty, alternatives, perspectives, and conditions for revision. For some decisions, the dossier may generate a score. But the score should be understood as local compression, not as the truth about the artifact.

LLM-as-a-judge research already provides concrete audit techniques. Position bias can be tested by swapping candidate order. Repetition stability can be measured by rerunning the same comparison. Preference fairness can examine whether equivalent candidates receive different treatment. Other studies document sensitivity to authority cues, gender information, stylistic polish, and misinformation. These perturbations should become standard unit tests for relevance evaluators in the same way software systems are tested for edge cases.

The system also needs calibrated abstention. When critics disagree strongly or evidence is thin, the correct output may be 'unresolved' rather than a forced number. Escalation rules can route such cases to domain experts or request additional evidence. This is especially important for novelty and cultural value, where uncertainty is often intrinsic rather than a temporary engineering defect.

A synthesis layer can still produce a decision, but it should preserve the structure underneath. It may apply hard vetoes for safety constraints, produce Pareto fronts for trade-offs, or calculate a weighted score when an institution explicitly defines weights. What it should not do is hide contestable assumptions inside a language model's latent preferences. The value of AI lies in making complex judgment more inspectable, not in making institutional values disappear.

10.3 Portfolio and Longitudinal Intelligence

Individual evaluation is only half the problem because organizations choose portfolios. A funder chooses a set of projects, a publisher a list of books, a platform a recommendation slate, and a design organization a roadmap. The marginal relevance of an item depends on what has already been selected. The eleventh project using the same technical architecture may add less value than the first project testing a credible alternative, even if its isolated score is slightly higher.

Portfolio selection can therefore use concepts from multi-objective optimization. Rather than collapsing every dimension into a weighted sum, the system can identify Pareto-efficient candidates, impose diversity or coverage constraints, reserve budgets for high-uncertainty exploration, and model dependencies among items. These methods make trade-offs explicit. They also permit an institution to say something more meaningful than 'we funded the top twenty scores': we funded a portfolio that covers several hypotheses, risk levels, communities, and time horizons.

Such a system should begin with an explicit schema, not merely with a prompt. For each evaluation, the context builder would produce an object containing the decision to be made, stakeholders, constraints, values, time horizon, and which criteria are gates versus trade-offs. The evidence layer would attach to each claim its sources and evidence type: paper, test, simulation, policy document, human preference, precedent, or critical argument. The model no longer vaguely "remembers" why it gave a score; the justification can be inspected and recomputed.

Longitudinal evaluation closes the loop that most assessment systems leave open. If a funder never compares ex-ante judgments with what later happened, it cannot learn which criteria were predictive, which reviewers were calibrated, or which kinds of novelty it systematically missed. The same is true for publishers, design systems, and cultural platforms. A relevance system should accumulate a memory of its own errors.

Such infrastructure also needs a governance layer. Relevance judgments allocate scarce attention, funding, publication opportunities, visibility, and sometimes legitimacy. The system should therefore record who defined the criteria, which evidence sources were admissible, which models were used, where human overrides occurred, and how an affected party can contest a factual or procedural error. Auditability is not an administrative afterthought; it is part of the epistemic design because unexplained decisions cannot be systematically corrected.

The most valuable long-term dataset would not be a collection of final scores. It would be a history of predictions and revisions. An evaluator might record that a project is likely to achieve a particular outcome under stated assumptions, that a novel appears derivative relative to a defined corpus, or that a design is robust only inside a specified operating envelope. Years later, the system can ask which assumptions failed, which signals were misleading, and which apparently weak evidence turned out to be important. This transforms evaluation from verdict production into institutional learning.

Relevance infrastructure is therefore closer to a constitutional layer for machine-assisted intellectual production than to a recommendation widget. It defines how reasons are represented, how uncertainty survives compression, how disagreement remains visible, how portfolios retain exploration, and how old judgments can be reopened. If AI makes generation abundant, this layer determines which possibilities receive the scarce resources needed to become real. Its quality will shape not only what machines create, but which parts of the possible future societies repeatedly choose to notice.

Specialized evaluators should have narrow mandates. A novelty evaluator searches for the closest neighbors and meaningful differences. A feasibility evaluator searches for dependencies, resources, and failure modes. A relevance-to-call evaluator maps claims to official requirements. A cultural critic searches for relations to tradition and communities. An adversarial critic attempts to falsify the arguments of the others. Separation of roles makes local benchmarks possible: we can measure whether the novelty detector misses precedents without confusing that error with a judgment about impact.

The aggregator should not be a simple weighted sum. For decisions with hard constraints, it can apply explicit vetoes. For trade-offs, it can preserve a Pareto frontier. For disagreement, it can report distributions or intervals. For portfolios, it can add diversity and complementarity. When two alternatives are practically indistinguishable within the error margin, it can return an

equivalence class rather than invent an ordering. This formal modesty is more valuable than three decimal places of false precision.

Finally, the system needs longitudinal memory. Every dossier should be reopenable as new data arrive: the project was funded, the design entered production, the book found an unexpected audience, the film became a cultural reference. Initial predictions are compared with outcomes and evaluators are recalibrated by dimension. Assessment thereby becomes not a scoring service, but a scientific programme about its own capacity to judge.

Longitudinal memory turns this architecture into a scientific instrument. Every dossier should remain linked to later outcomes. Funded proposals generate deliverables, papers, software, adoption, and sometimes failure. Recommended films generate viewing patterns and cultural traces. Selected designs generate use data and maintenance problems. The system can compare predicted relevance with observed trajectories by dimension, revealing where its evaluators are calibrated and where they are systematically wrong.

This feedback loop should include false-negative research whenever possible. Near-threshold rejected proposals can be tracked through later publications or alternative funding. Unselected designs can sometimes be tested in controlled studies. Cultural recommendation systems can deliberately expose a small random sample of lower-ranked works to estimate what the ranking would otherwise hide. Exploration is therefore not only a value choice; it is a method for collecting counterfactual data about the evaluator.

The mature product is not an AI that pronounces relevance. It is an infrastructure that makes relevance claims explicit, challenges them, records uncertainty, supports portfolio decisions, and learns from later consequences. Such a system can be useful precisely because it is modest about what it knows. Its intelligence lies less in the authority of a score than in the quality of the questions it forces an institution to answer.

The deepest consequence of generative abundance is that production no longer proves commitment. In older cultural and scientific systems, creating a book, film, dataset, or research proposal required enough scarce labor and capital that production itself carried a weak signal: someone had decided the object was worth substantial effort. That signal was never reliable, and it excluded many capable people, but it limited volume. When AI reduces the cost of producing credible artifacts, existence loses much of that filtering meaning.

New institutions will therefore emerge around certification of evaluation. A research result may arrive with provenance, replication status, and an evidence graph. A design may arrive with simulation results and known failure modes. A book may carry an originality analysis, source map, and record of critical review. A film may be accompanied by audience evidence and a statement of how it contributes to portfolio diversity. These additions do not make the artifacts objectively valuable. They make the process by which attention is allocated more legible.

An economy of abundant creation changes the meaning of scarcity. In the previous century, we could often assume that a published book had passed through a costly filter, that a film consumed enough resources for producers to perform selection before shooting, and that a serious research proposal required months of work. Those costs did not guarantee quality, but they limited volume. AI removes part of that constraint.

When the marginal cost of an intellectual artifact approaches zero, the fact that it "was produced" ceases to carry much signaling value. We will need institutions that certify not only authenticity or provenance, but also what kind of evaluation an artifact has undergone. A paper may arrive with an evidence graph and replication status. A design with simulations and failure modes. A book with source mapping, comparison with the literature, and originality assessment. A film with audience profiles but also arguments about cultural diversity. None of these guarantees value; they are infrastructures for judgment.

There is, however, a symmetrical danger. If relevance becomes too efficient, we may destroy exactly what we do not yet know how to value. Every selection system learns from the past, and the past contains canon, reputation, power, unequal distributions, and criteria that favored dominant forms. An evaluator perfectly calibrated to historical relevance could become an adversary of future relevance.

A mature system must therefore preserve a budget for the unknown. In research we call it high-risk/high-gain, frontier research, or exploratory funding. In recommendation we call it serendipity and exploration. In culture we can call it space for work that is difficult to classify. The principle is the same in every case: do not fully optimize the current distribution of preferences and criteria.

The danger is overcorrection. If we build extremely efficient relevance filters, we may eliminate exactly the artifacts that current models cannot understand. Historical data are full of prestige hierarchies, market inequalities, disciplinary fashions, and canons produced under unequal access. A model optimized to reproduce historical success can become an adversary of future novelty. The problem is especially serious when the system is used upstream, before an idea receives the resources required to prove itself.

For that reason every serious relevance institution needs an explicit exploration reserve. Frontier research calls call it high-risk/high-gain. Recommender systems call it exploration or serendipity. Investors call it optionality. Archives call it preservation. The names differ, but the function is the same: protect some possibilities from the full force of current optimization. The reserve should not be a romantic refuge from evidence; it should be a structured acknowledgment that the evaluator's model of value is incomplete.

Selection under abundance is therefore not merely ranking. It is a constitutional problem for knowledge and culture. We are deciding which questions are tested, which voices are amplified, which designs become infrastructure, which stories become shared

reference points, and which alternatives remain available when the dominant path fails. Relevance systems will quietly govern these choices whether or not we give the field a name.

A science of relevance should begin by treating evaluators as hypotheses. Instead of asking whether a model agrees with today's judges, we should ask what construct it measures, how stable it is under perturbation, how it behaves across communities, whether it predicts later outcomes, and what classes of valuable objects it systematically misses. Agreement with current consensus is only one validation target, and for frontier novelty it may be a dangerous one.

The first research programme is longitudinal grant evaluation. Proposal scores, review text, reviewer characteristics, funding decisions, outputs, adoption, and later impact should be connected under strong privacy controls. Baselines based on applicant reputation, institution, budget, and field can be compared with richer relevance dossiers. The central question is whether multidimensional, evidence-linked evaluation predicts outcomes better while reducing known biases. The recent ERC homophily study and the 2026 GenAI-novelty result show that there is already enough signal to formulate falsifiable hypotheses.

Public benchmark design will be decisive. Useful datasets should include not only obvious winners and losers but near-threshold decisions, delayed successes, negative results, unconventional proposals, minority viewpoints, and cases in which expert communities disagree. Benchmark labels should preserve distributions and rationales rather than a single gold score whenever possible. Otherwise the benchmark will teach machines to reproduce the very oversimplification the field is trying to escape.

The real objective is not to build an AI that knows what is relevant. It is to build systems that make explicit why we consider something relevant, for whom, under which conditions, and with what uncertainty; systems whose judgments can be challenged, compared, and corrected. Relevance is less a property than a formalized conversation about value.

This may become one of the central disciplines of the AI era. If artificial intelligence can create almost anything, progress will not depend only on the capacity to generate more. It will depend on our ability to see what deserves continuation, what deserves protection from metrics, what deserves testing, what deserves to be shown to others, and what deserves to wait until the world becomes capable of understanding it.

A science of relevance would therefore have an unusual object: not only artifacts, but also the systems that select them. It would study error distributions, how criteria change creator behavior, how portfolios shift, and what is lost through false negatives. In many domains, the most expensive failure is not selecting a mediocre idea but systematically eliminating classes of ideas that do not resemble the past. Benchmarks should therefore include cases of delayed recognition and minority ideas, not only contemporary consensus.

In a world of abundant intellectual production, pluralism is no longer only a moral value; it is a strategy of resilience. Multiple evaluators, multiple criteria, and an exploration budget keep options open in an environment whose future cannot be fully predicted. Relevance becomes a discipline of selection under uncertainty: rigorous enough to remove noise, modest enough not to remove the future.

The second programme is cross-domain transfer. Components such as novelty retrieval, provenance tracking, uncertainty estimation, adversarial critique, portfolio optimization, and longitudinal calibration can be tested across research, design, literature, and film. The goal is to discover which mechanisms transfer and which fail because the construct changes. A novelty detector may transfer well; a utility score probably does not. This is where a general architecture can be empirically separated from a universal scoring function.

The third programme is Goodhart stress testing. Generator agents should be allowed to see the evaluator and iteratively optimize against it. Researchers can then measure how quickly the proxy degrades, which artifacts exploit it, and whether ensembles or

hidden tests improve robustness. The fourth programme is false-negative discovery: deliberately sample rejected or low-ranked items and measure whether later evidence reveals missed value. The fifth is institutional portfolio evaluation, where success is measured not only by average item quality but by diversity, resilience, breakthrough rate, and coverage of strategic objectives.

The commercial opportunity follows directly from the research agenda, but it should be approached as infrastructure rather than oracle-building. Universities, funders, publishers, venture studios, media platforms, design organizations, and governments all face the same abundance problem in different forms. A trustworthy relevance platform would provide context specification, evidence retrieval, specialized critics, bias audits, portfolio analysis, and longitudinal learning while keeping the institution's values explicit. If AI can create almost anything, one of the most consequential technologies of the next decade may be the machinery that helps us decide what deserves another experiment, another reader, another euro, another hour of attention - and what deserves to remain possible even before we understand why.

The central proposal is intentionally anti-oracular. A relevance system should help institutions see more clearly what they are deciding, which evidence supports the decision, where uncertainty remains, what alternatives were excluded, and what future observations would justify revision. Its intelligence lies as much in preserving disagreement as in producing recommendations.

When creation becomes abundant, civilization does not mainly need machines that can make more things. It needs institutions - human, computational, and hybrid - that can decide what deserves scarce attention without systematically deleting the unfamiliar future. That is the durable problem of relevance.

APPENDIX A. CONCEPTS AND TERMS

This appendix explains concepts that may be familiar to specialists but should not be required background for a general reader. The explanations are deliberately plain-language. The linked Wikipedia pages are entry points rather than authoritative sources; the academic and institutional bibliography that follows should be used for substantive claims.

Term	Plain-language explanation and accessible link
Altmetrics	Alternative metrics that track attention to research outside traditional citations, for example online mentions or saves. Wikipedia
Bibliometrics	Quantitative analysis of publications and citations. Useful for mapping research activity, but not a direct measurement of truth or quality. Wikipedia
Calibration	The degree to which stated probabilities or confidence levels match observed frequencies over time. Wikipedia
Causal inference	Methods for reasoning about whether one factor actually contributes to an outcome rather than merely correlating with it. Wikipedia
Counterfactual	A structured question about what would have happened under a different condition, used to reason about causality and alternatives. Wikipedia
Epistemology	The study of knowledge: what counts as justified belief, evidence, and reliable ways of knowing. Wikipedia
Exploration-exploitation dilemma	The trade-off between trying uncertain alternatives that may teach us something and choosing options already known to perform well. Wikipedia

Term	Plain-language explanation and accessible link
External validity	The extent to which results observed in one study or context can be expected to hold in other relevant contexts. Wikipedia
Filter bubble	A proposed effect in which personalization reduces exposure to information or cultural material outside a user's established pattern. Wikipedia
Goodhart's law	The idea that a measure can become a poor measure once people or machines start optimizing directly for it. Wikipedia
Homophily	The tendency for people to associate with, prefer, or evaluate more positively others who share characteristics with them. Wikipedia
Multi-criteria decision analysis	A family of methods for comparing alternatives when several objectives or criteria matter at the same time. Wikipedia
Novelty	Difference from relevant prior work or artifacts. Novelty alone does not imply usefulness, truth, or importance. Wikipedia
Open science	Practices intended to make scientific processes and outputs more accessible, transparent, reusable, and reproducible. Wikipedia
Option value	The value of keeping a possibility available because future information may make it more useful than it appears today. Wikipedia
Pareto frontier	The set of alternatives for which improving one objective would require worsening at least one other objective. Wikipedia
Peer review	Evaluation of research or proposals by people with relevant expertise. Peer review supplies expert judgment but also contains disagreement and bias. Wikipedia

Term	Plain-language explanation and accessible link
Provenance	Information about where an artifact, datum, claim, or decision came from and how it was transformed. Wikipedia
Recommender system	A computational system that selects or ranks items for users based on predicted relevance, preference, or other objectives. Wikipedia
Scientometrics	The quantitative study of science, including publications, citations, institutions, collaboration, and research dynamics. Wikipedia
Technology readiness level	A scale used to describe how mature a technology is, from early principles to demonstrated operational systems. Wikipedia
Uncertainty quantification	Methods for representing and analysing uncertainty in models, measurements, predictions, and decisions. Wikipedia

Table A1. Concepts used in the book, explained for non-specialist readers.

The glossary is intentionally separate from the main chapters so that technical vocabulary does not interrupt the argument. A reader who already knows these terms can ignore the appendix; a student encountering them for the first time can return here without being forced to infer meaning from context.

Several terms have multiple technical definitions. Where that matters in the book, the chapter provides the narrower meaning being used. The links are therefore orientation aids, not substitutes for the cited research literature.

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